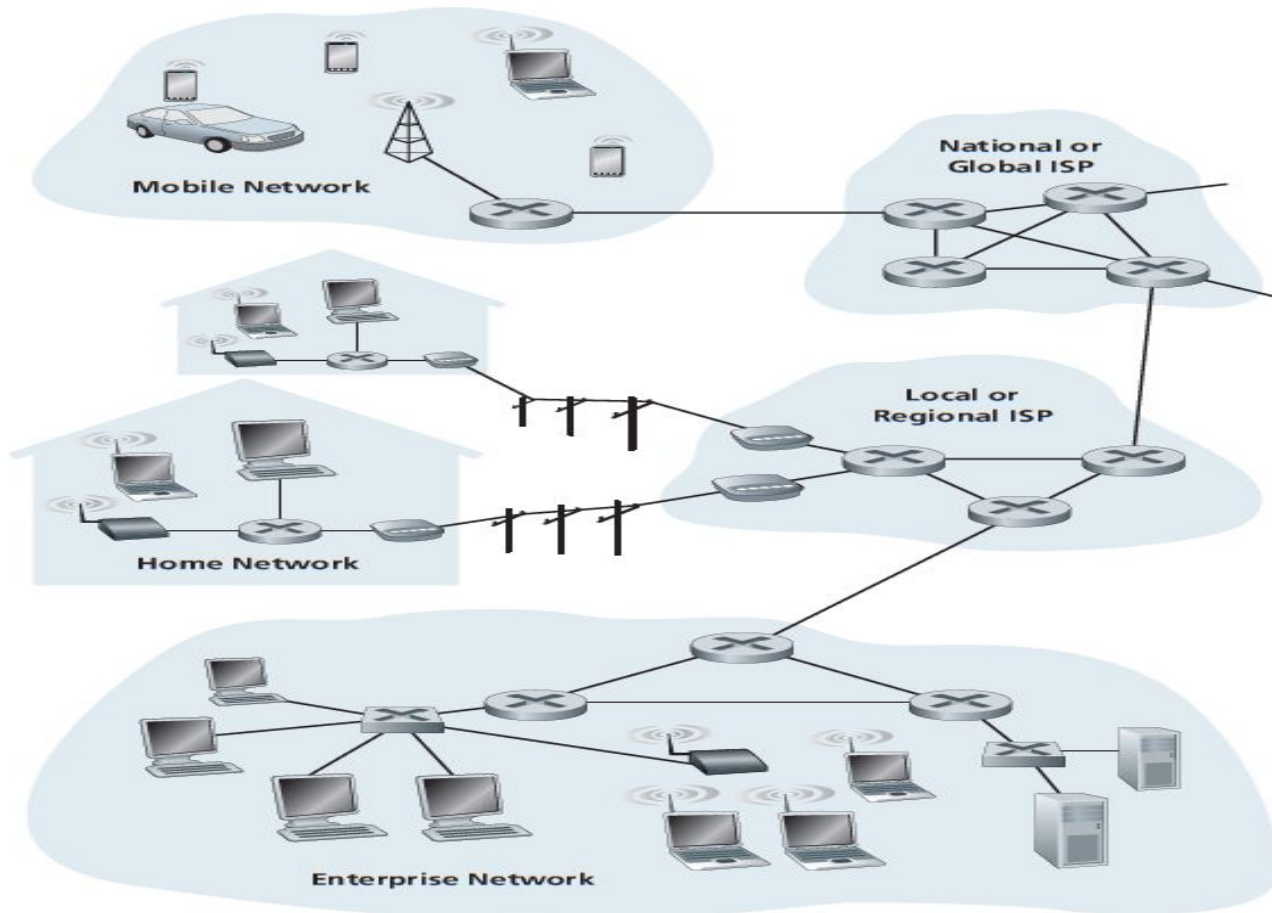


CSE 4615:: Wireless Networks

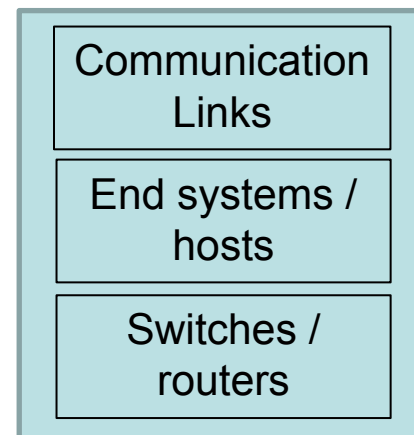
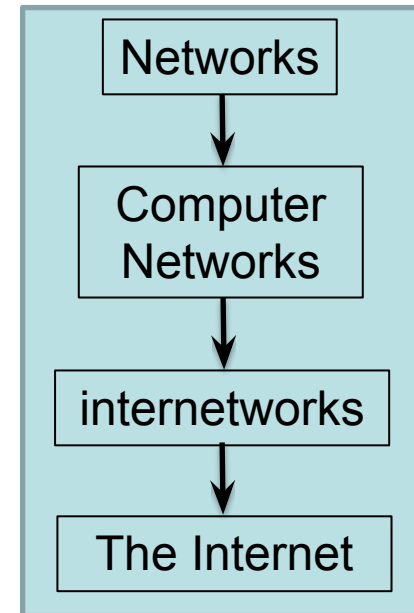
Overview of Wireless Access Networks

Lecture 2

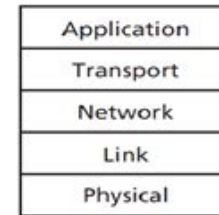
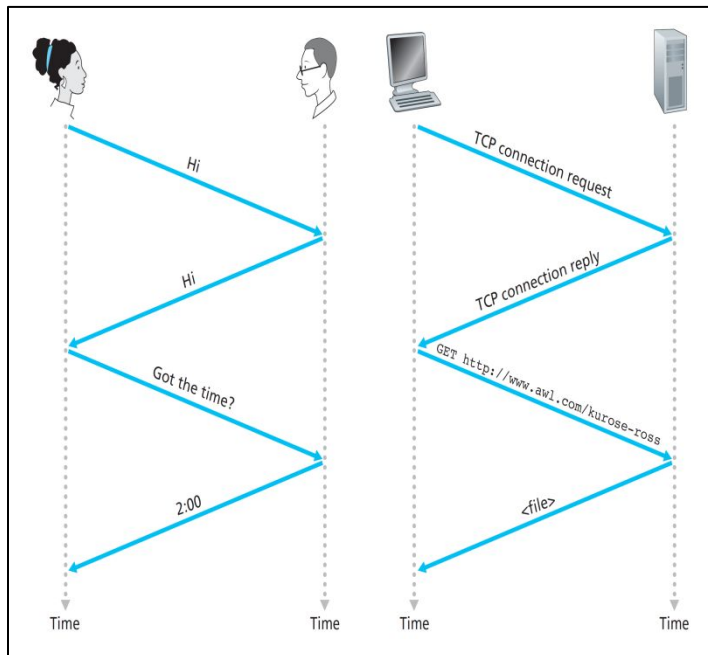
Internet – the bigger picture



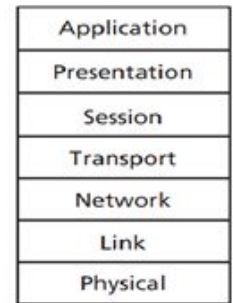
Key:



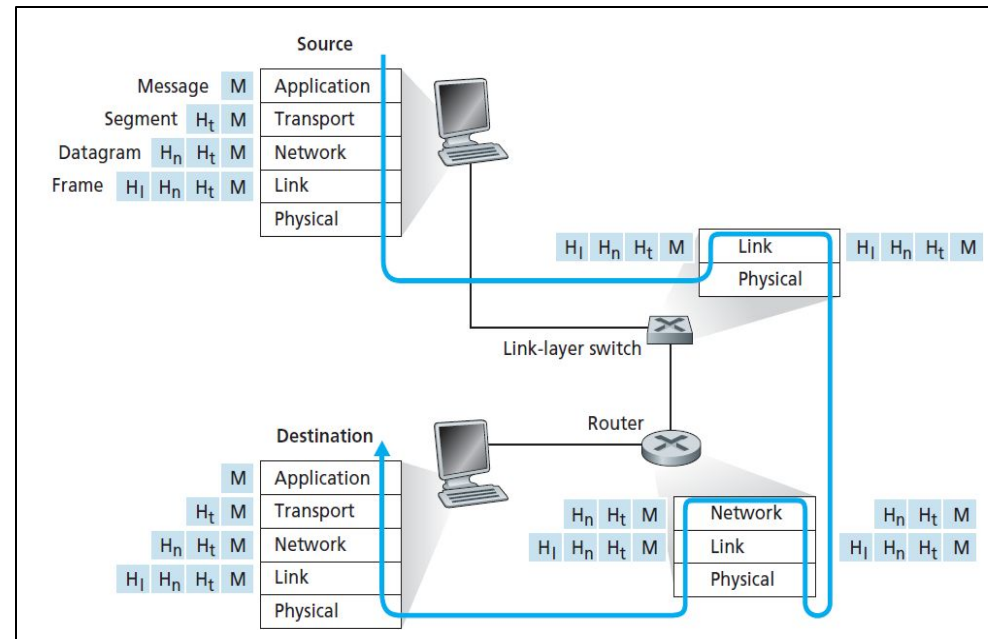
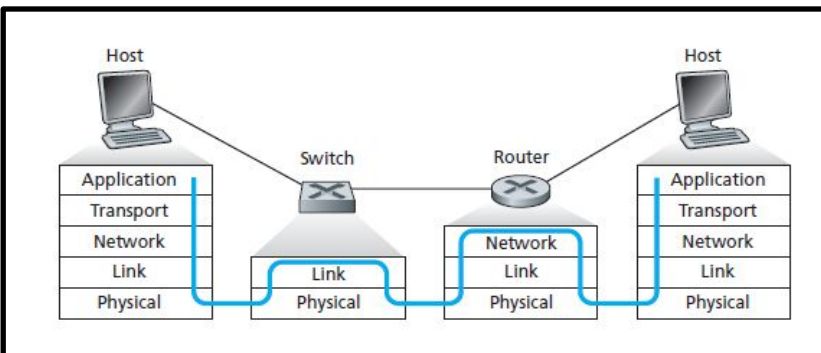
Protocols and Standards



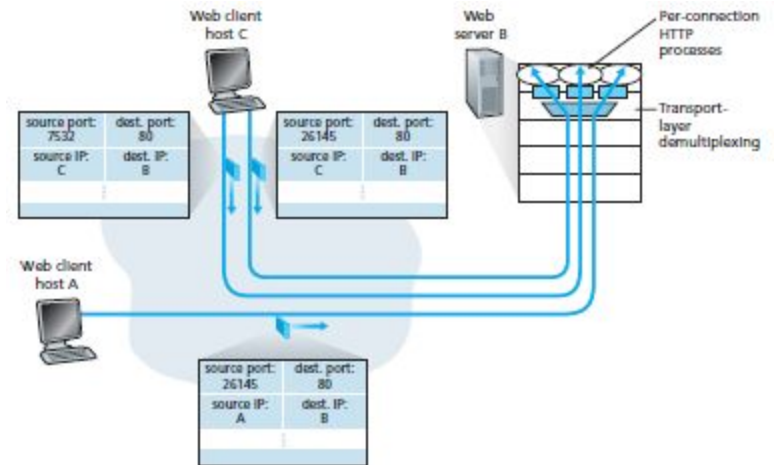
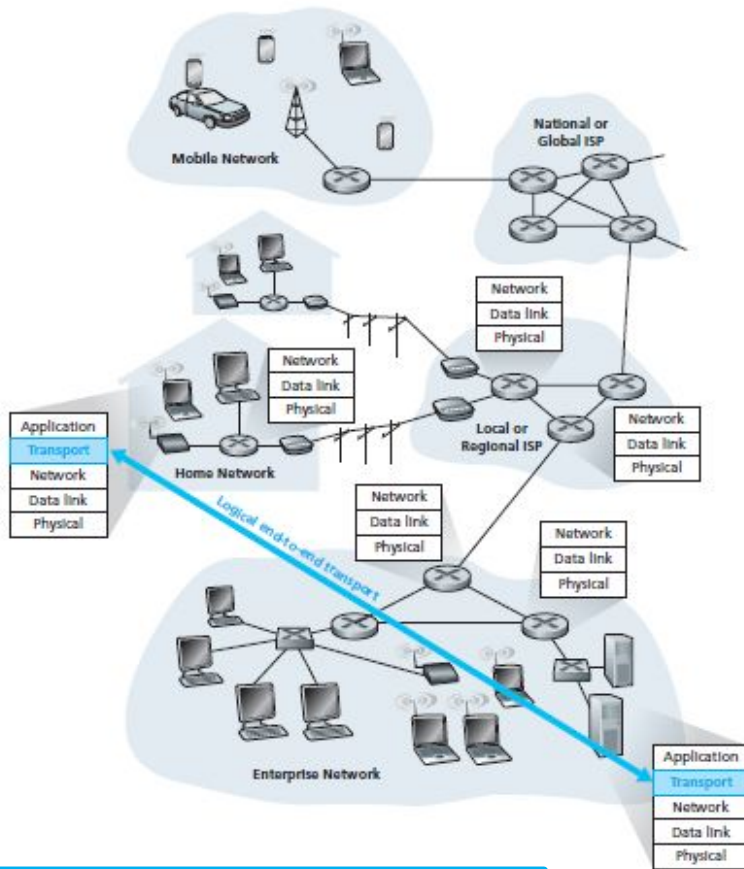
a. Five-layer Internet protocol stack



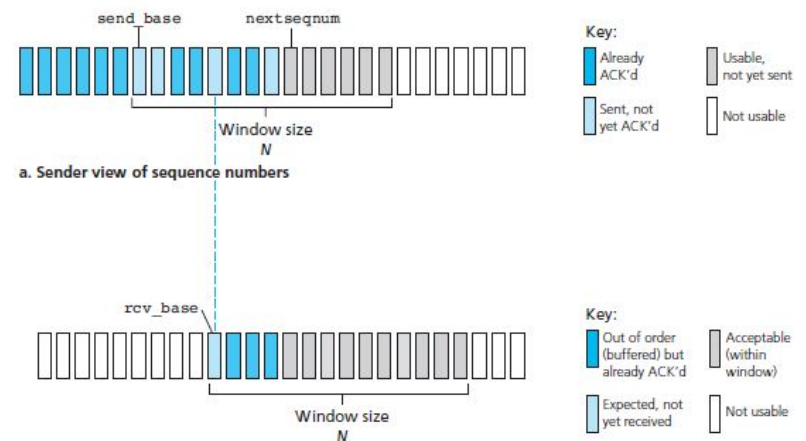
b. Seven-layer ISO OSI reference model



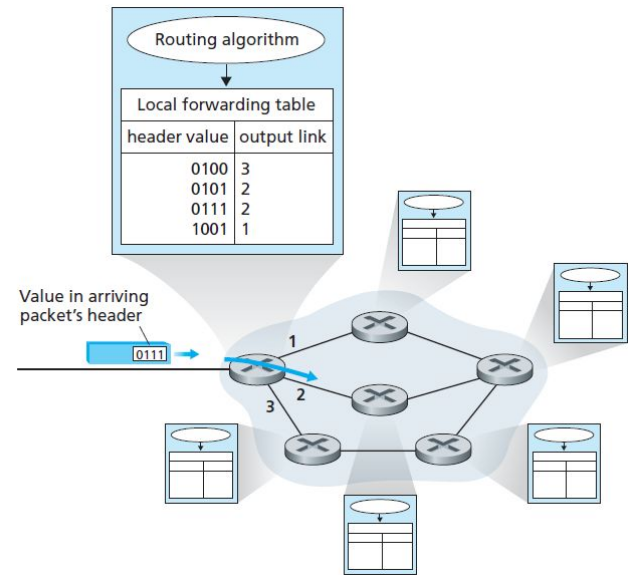
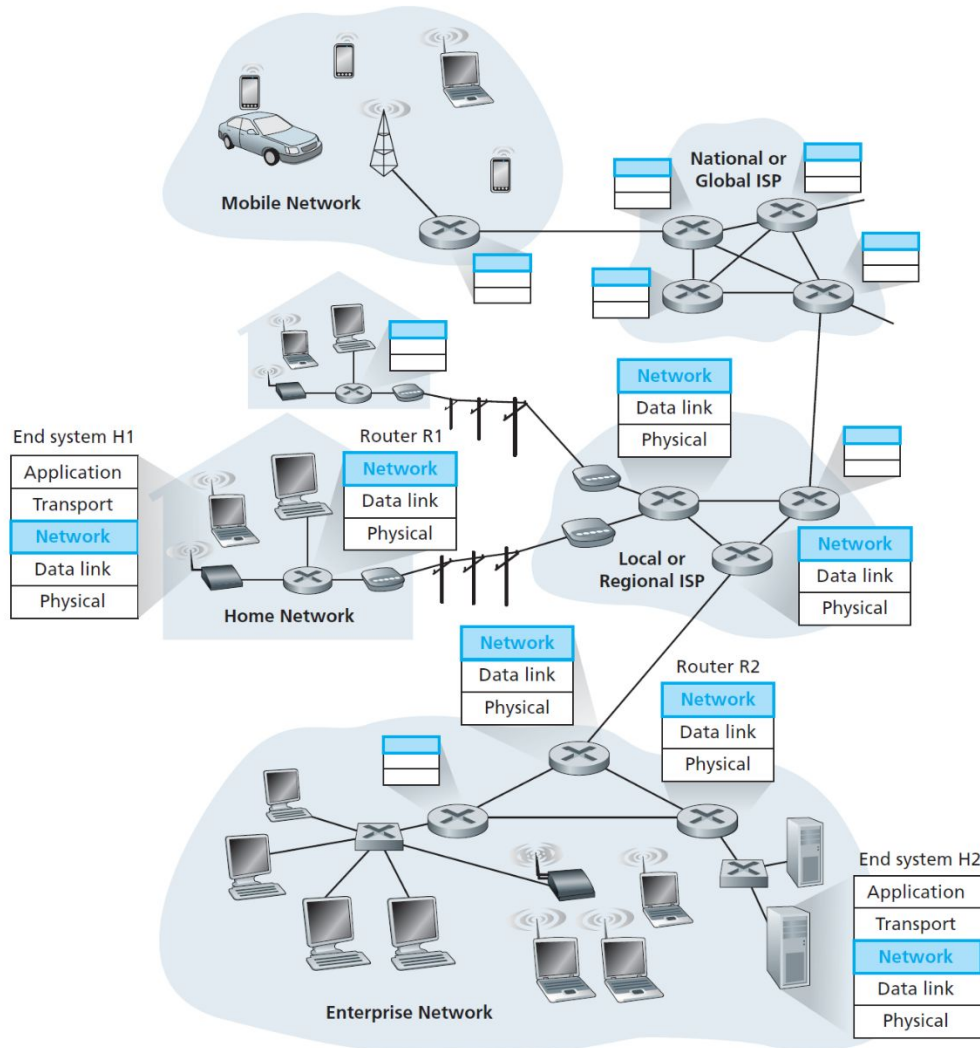
The Transport Layer



- Multiplexing / de-multiplexing
- Reliability & in-order delivery
- Congestion control
- Flow Control



The Network Layer

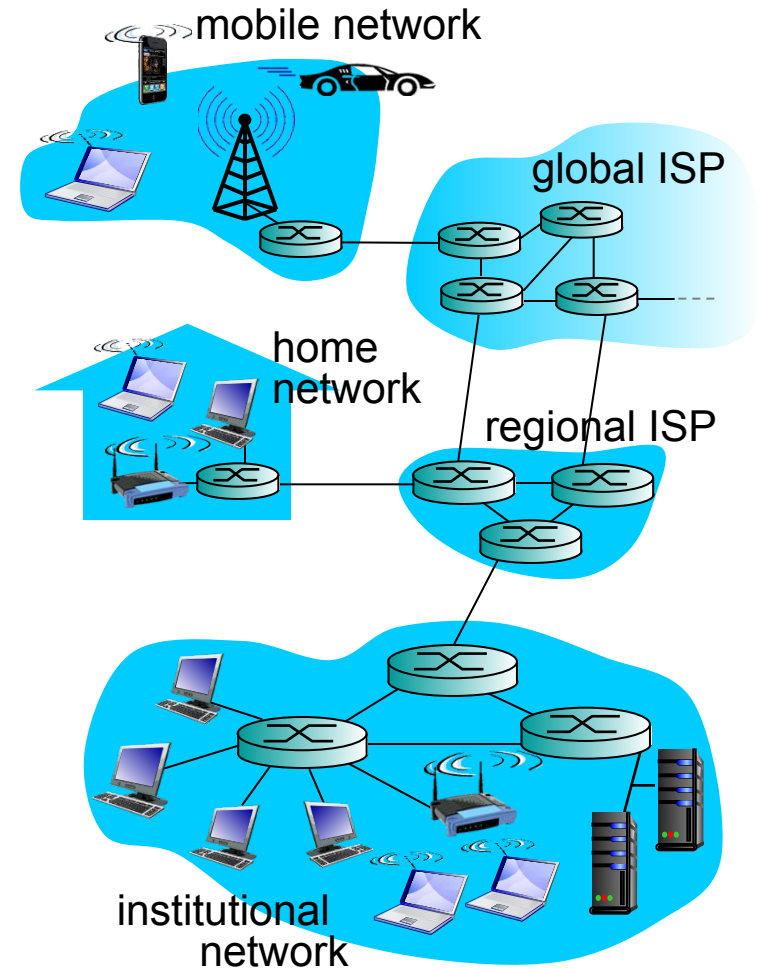


- Routing
 - Routing Metric
 - Hop-count, BW
 - ETX, ETT
- Routing Protocol
 - RIP, OSPF
 - BGP
- Packet Forwarding
- QoS

A closer look at network structure

A closer look at network structure:

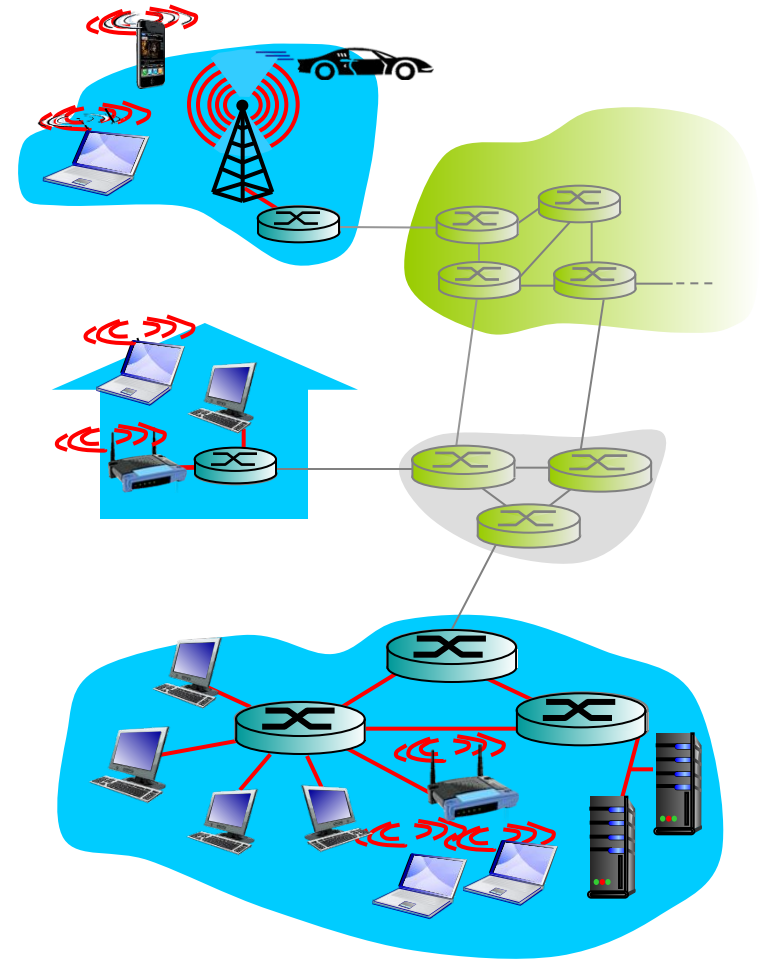
- *network edge:*
 - hosts: clients and servers
 - servers often in data centers
- ❖ *physical media:* wired, wireless communication links
- ❖ *network core:*
 - interconnected routers
 - network of networks



Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks



Access Networks

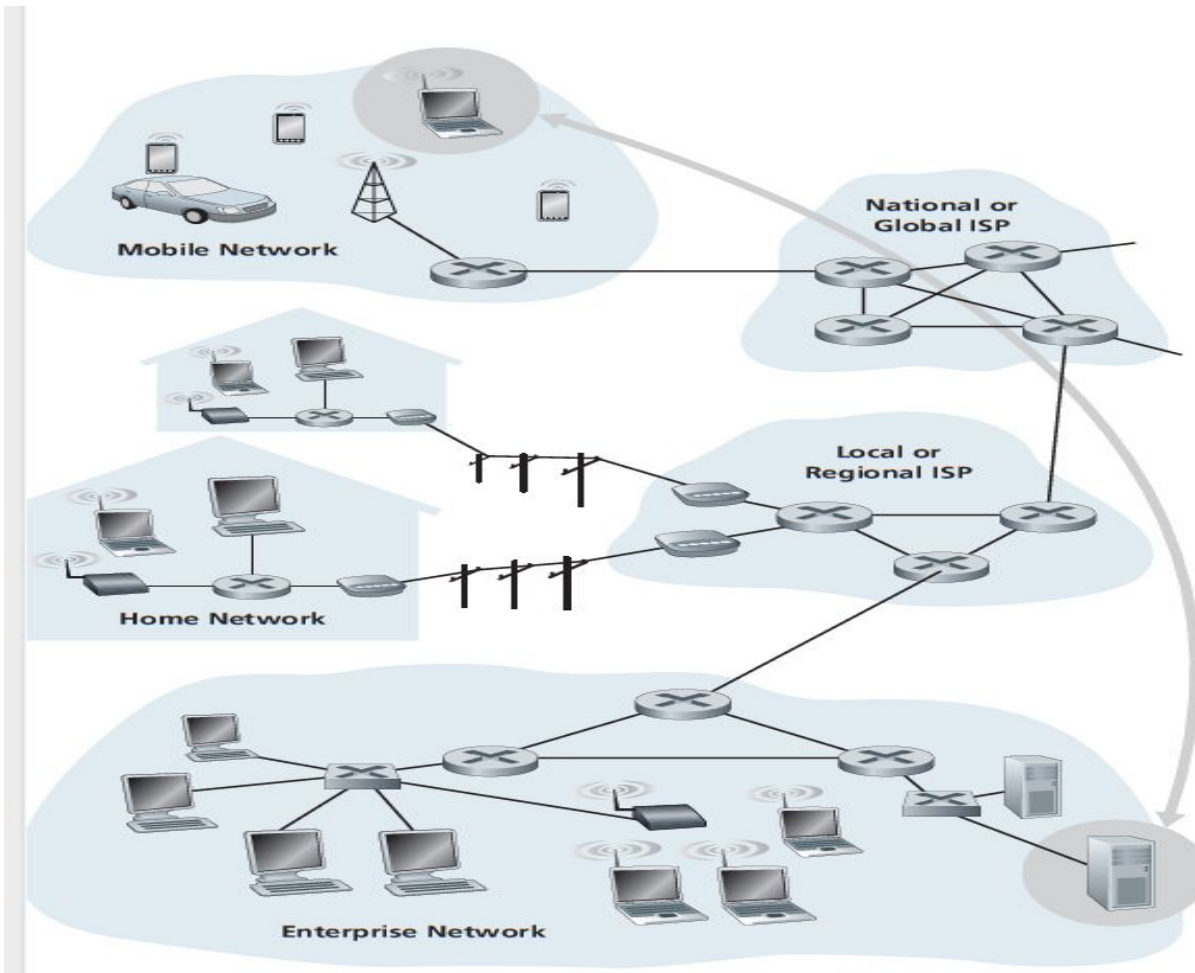
Access Networks

Nuts and Bolts of Internet



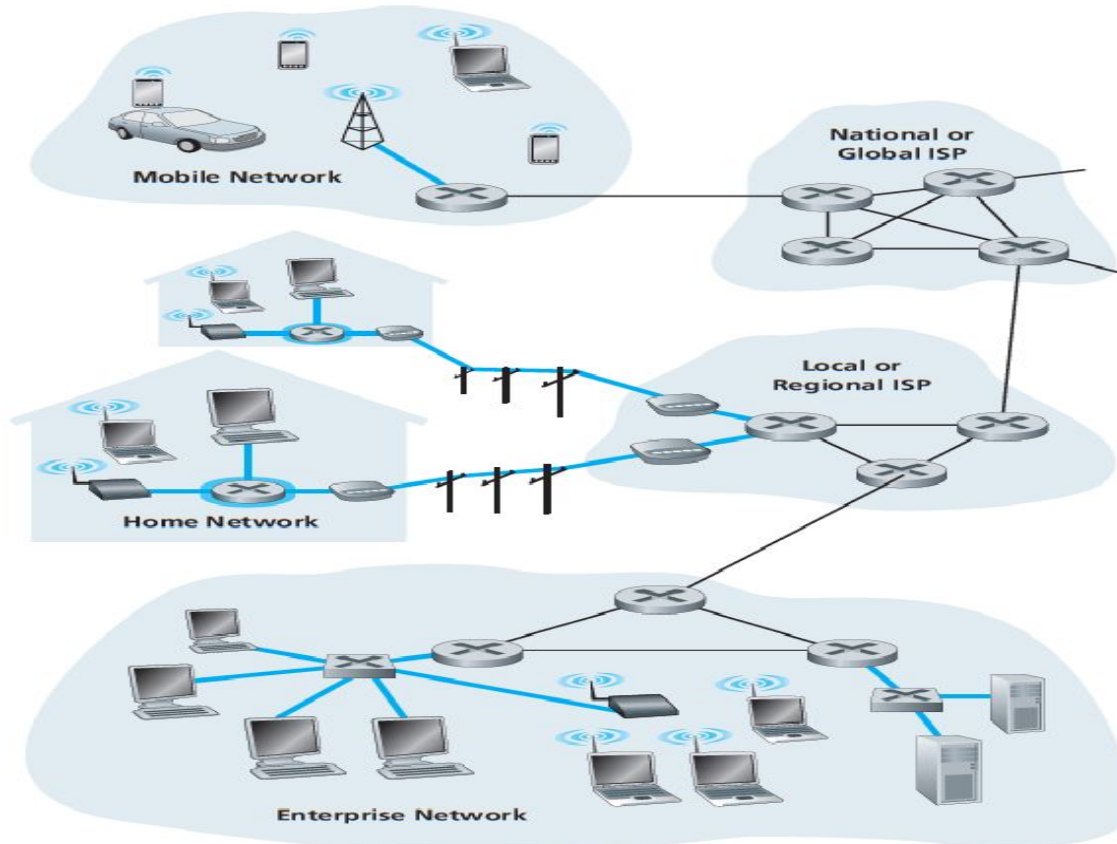
Access Networks

End System Interaction



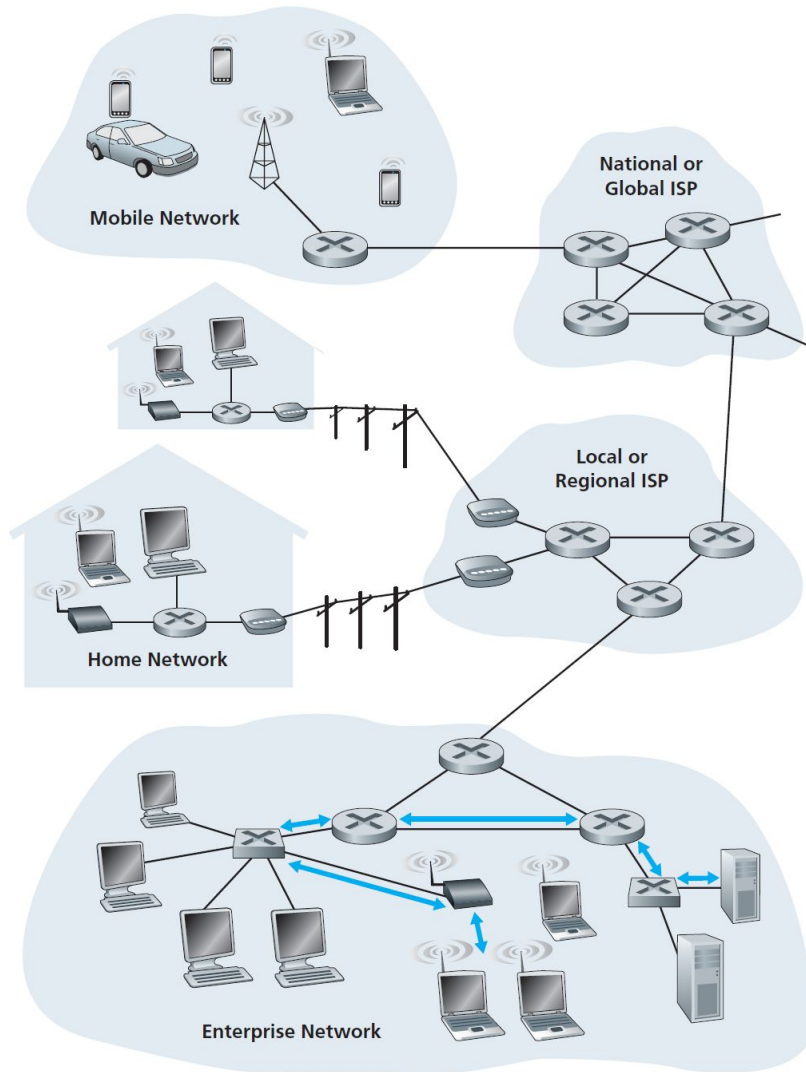
Access Networks

The network that **physically connects an end system to the first router** (also known as the “edge router”) on a path from the end system to any other distant end system



Access Networks

Taxonomy based on Technology Used



- Access Networks
 - Ethernet
 - WiFi-Based Access Networks
 - Wireless LAN
 - 802.11a/b/g
 - Ad-hoc Networks
 - Mesh Networks
 - DTN/ICN
 - VANET
 - 6LowPANs
 - WSNs
 - WiMAX-Based Access Net
 - Cellular Networks
 - 3G Cellular Networks and beyond
 - etc...

Access Networks

Taxonomy based on Technology Used

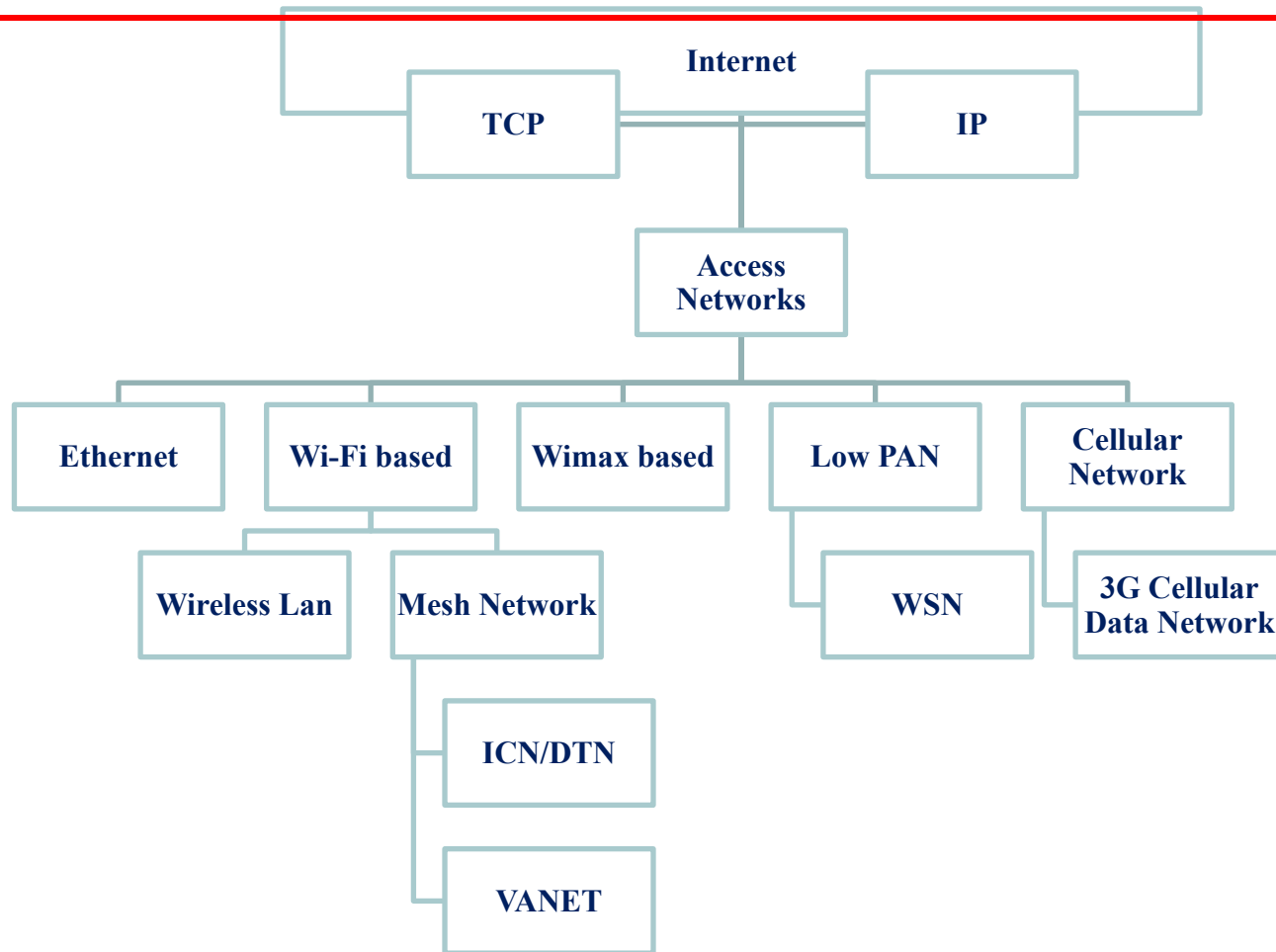
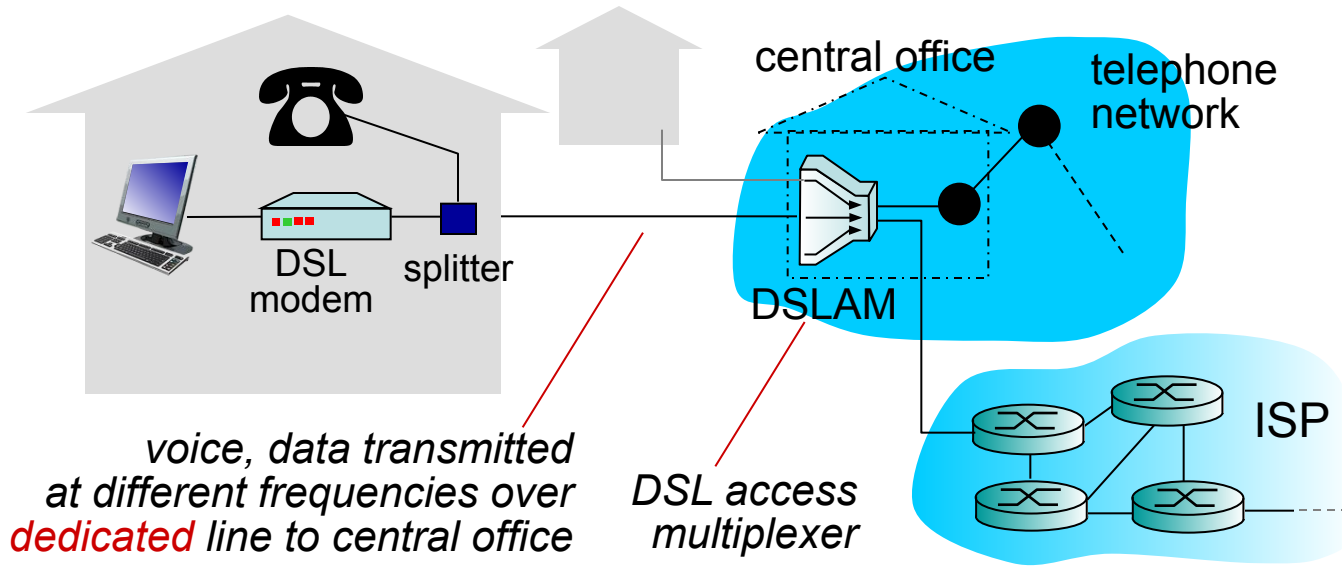


Figure: a taxonomy of access networks (approximate)

Access Networks

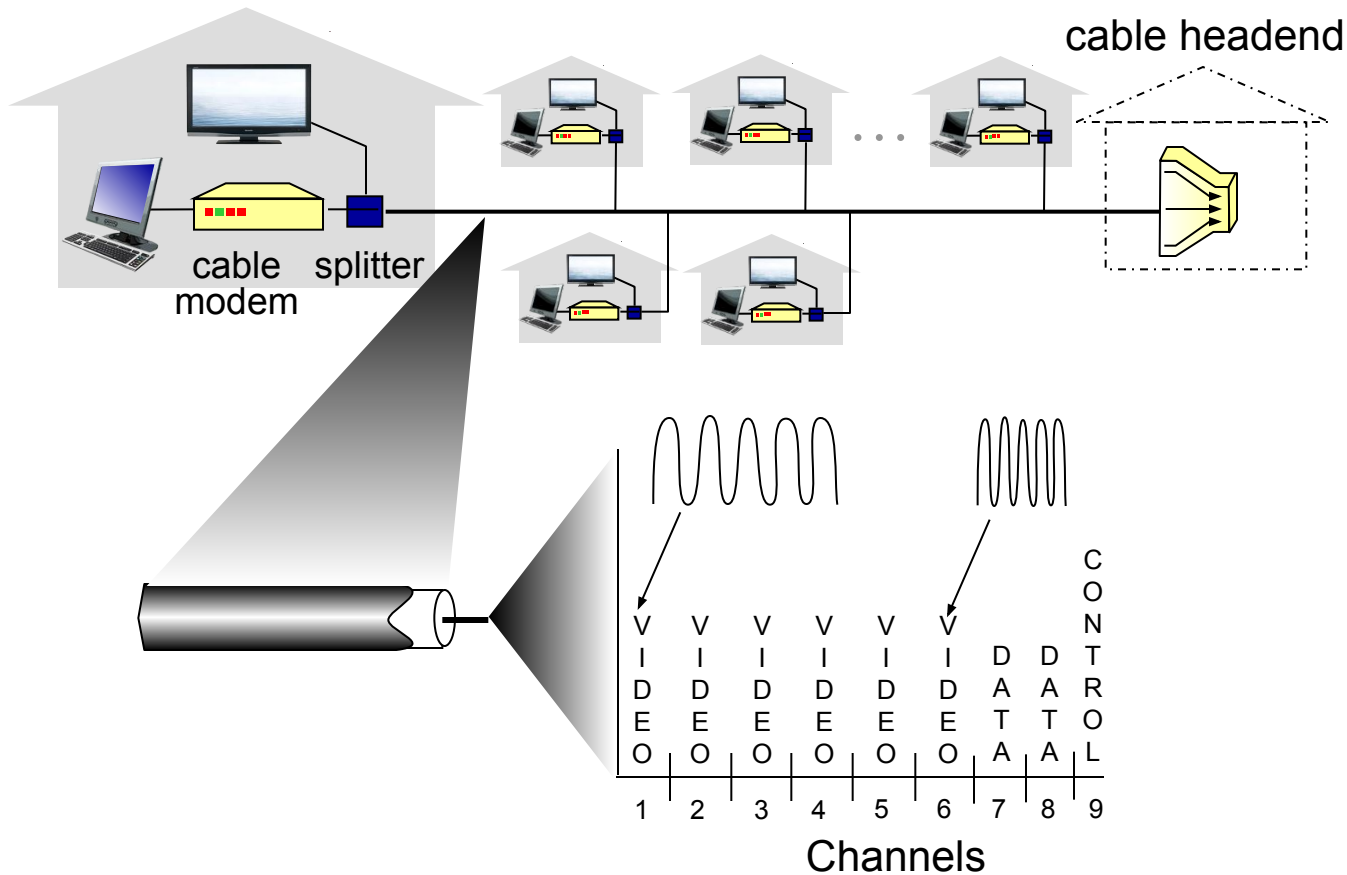
Wired Access Networks

Access net: digital subscriber line (DSL)



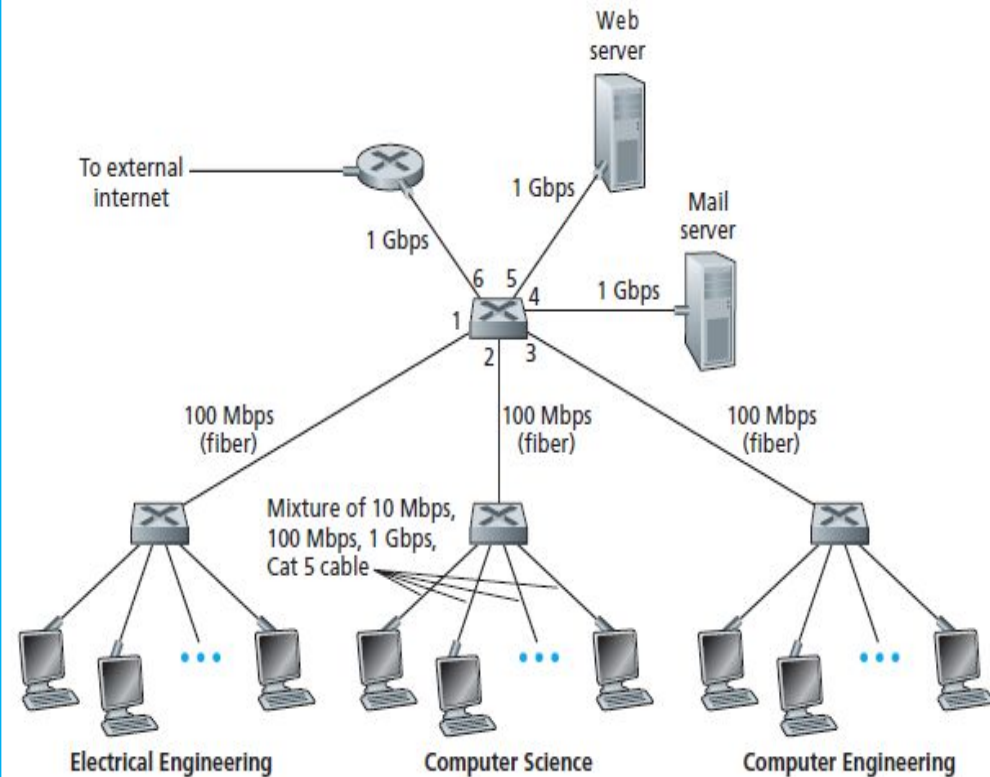
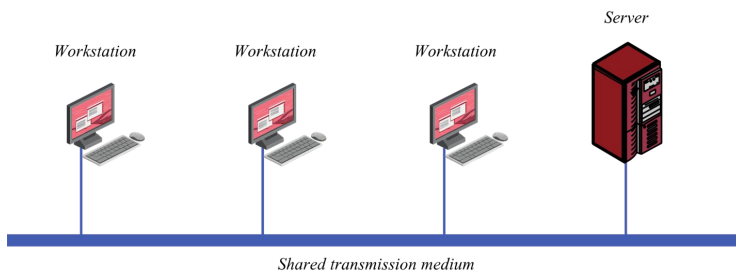
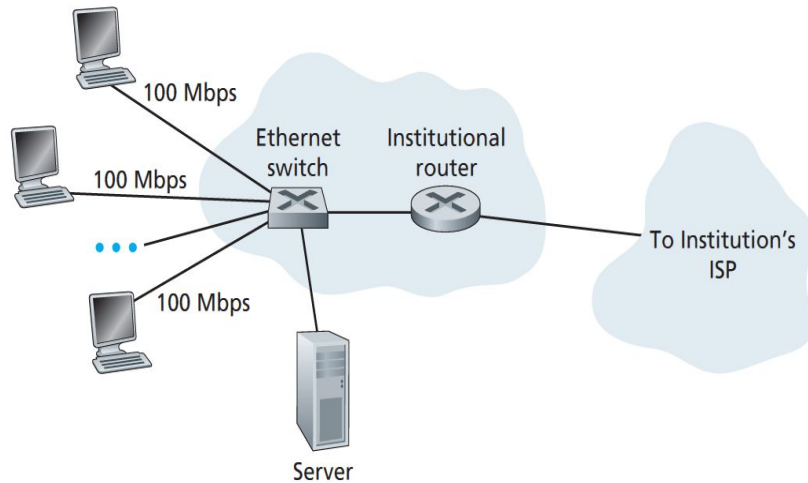
- ❖ use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- ❖ < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- ❖ < 24 Mbps downstream transmission rate (typically < 10 Mbps)

Cable Network



frequency division multiplexing: different channels transmitted in different frequency bands

Ethernet – Traditional Wired Access



Access Networks

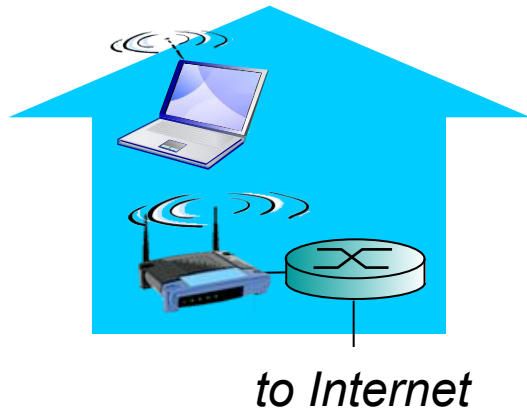
Wireless Access Networks

Wireless Access Networks

- shared *wireless* access network connects end system to router
 - via base station aka “access point”

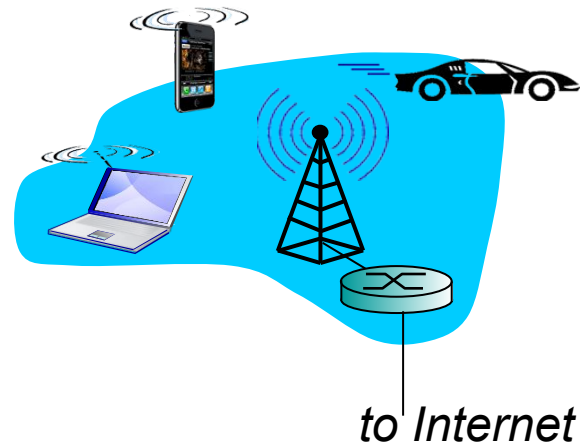
wireless LANs:

- within building (100 ft)
- 802.11b/g (WiFi): 11, 54 Mbps transmission rate

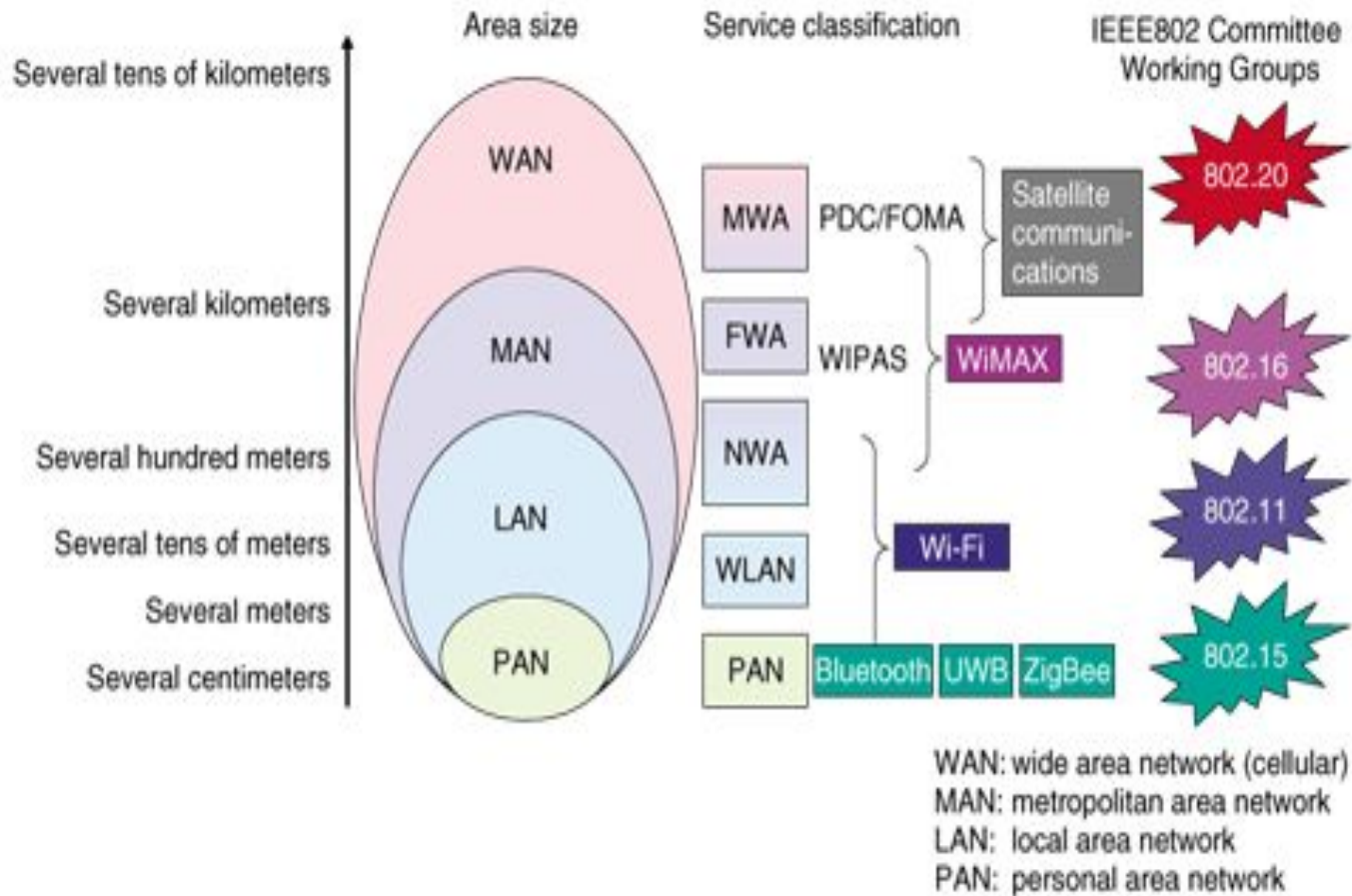


wide-area wireless access

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE

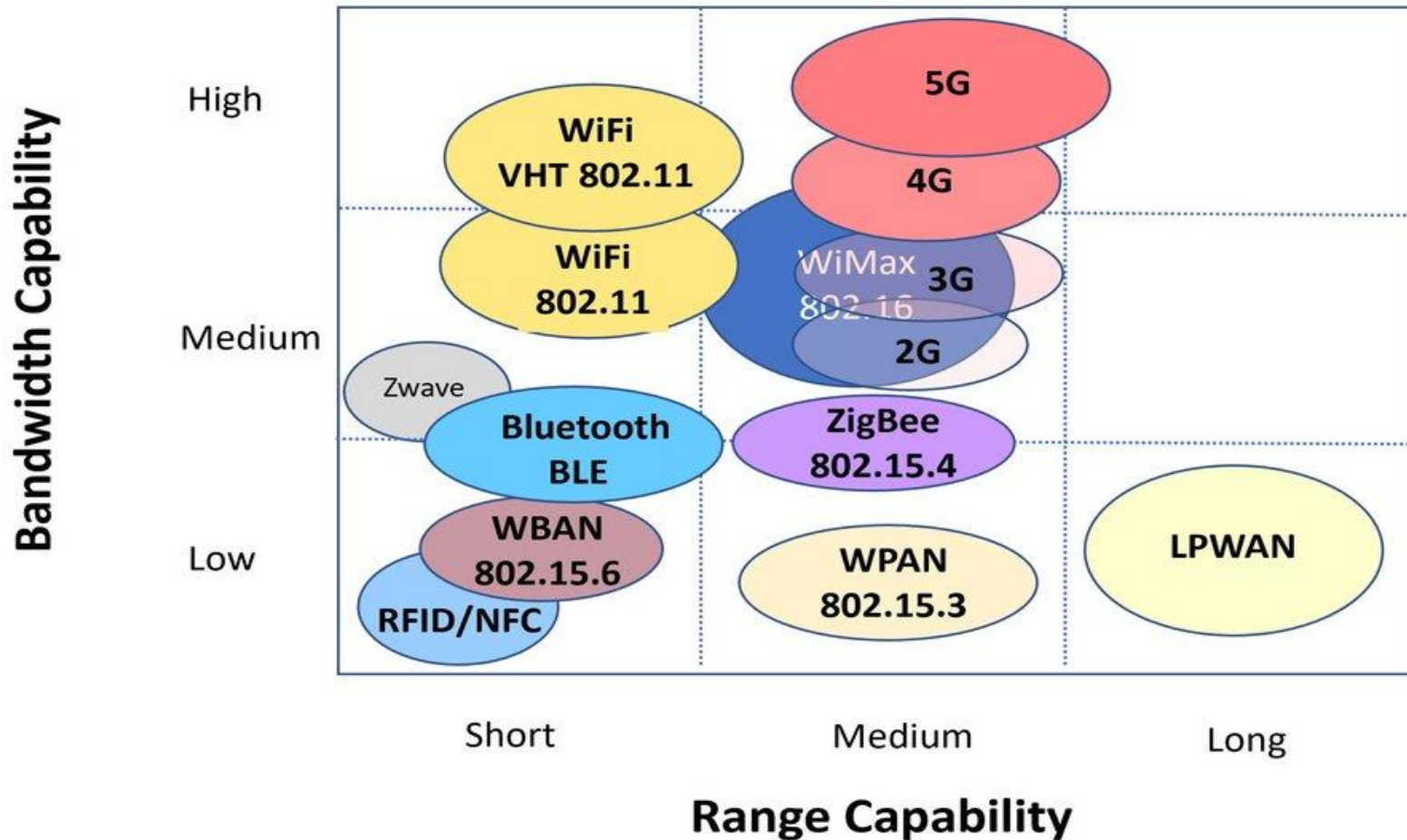


Wireless Access Networks



Wireless Access Networks: Range Comparison

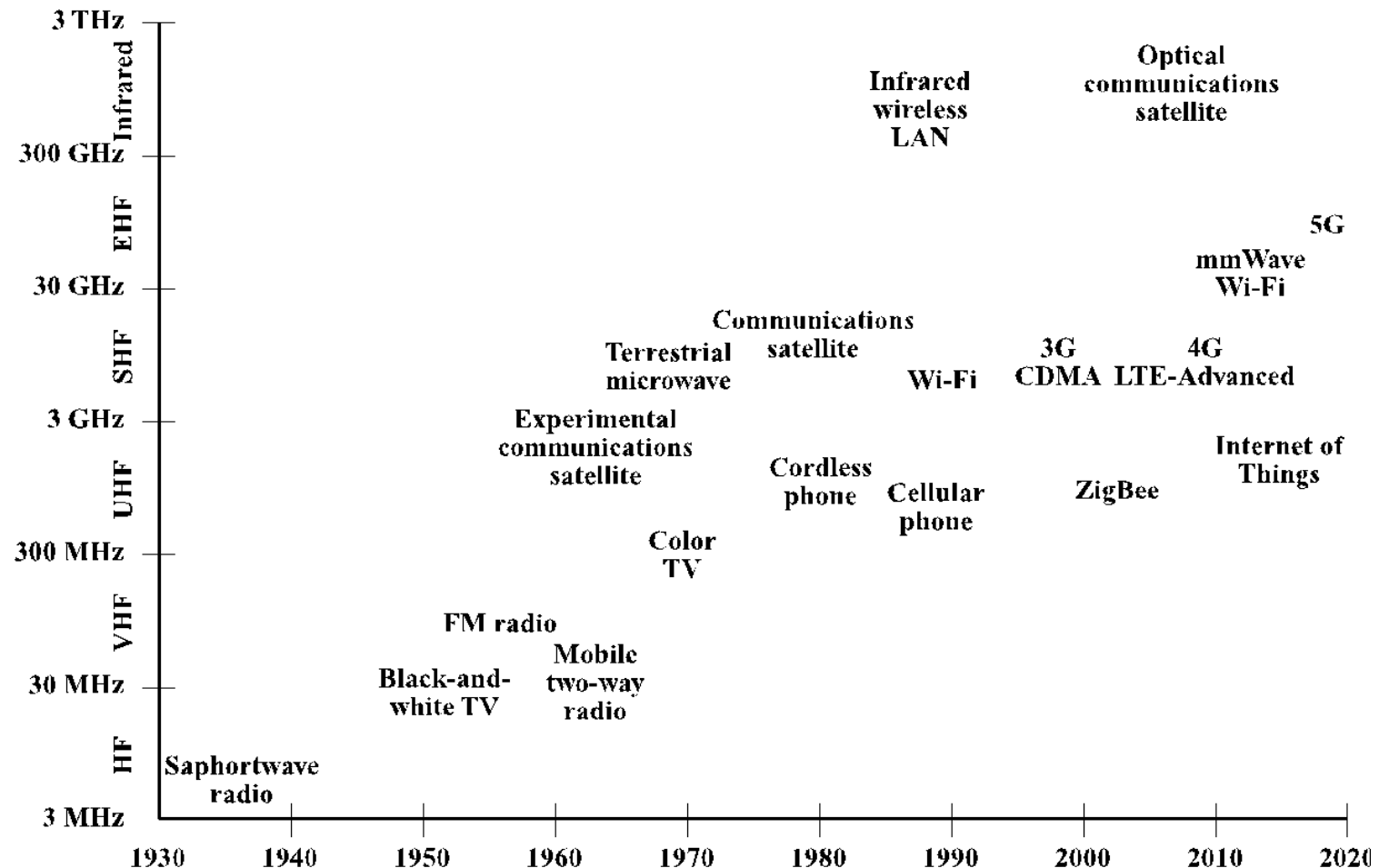
Comparison of Wireless Technologies



Wireless Access Networks

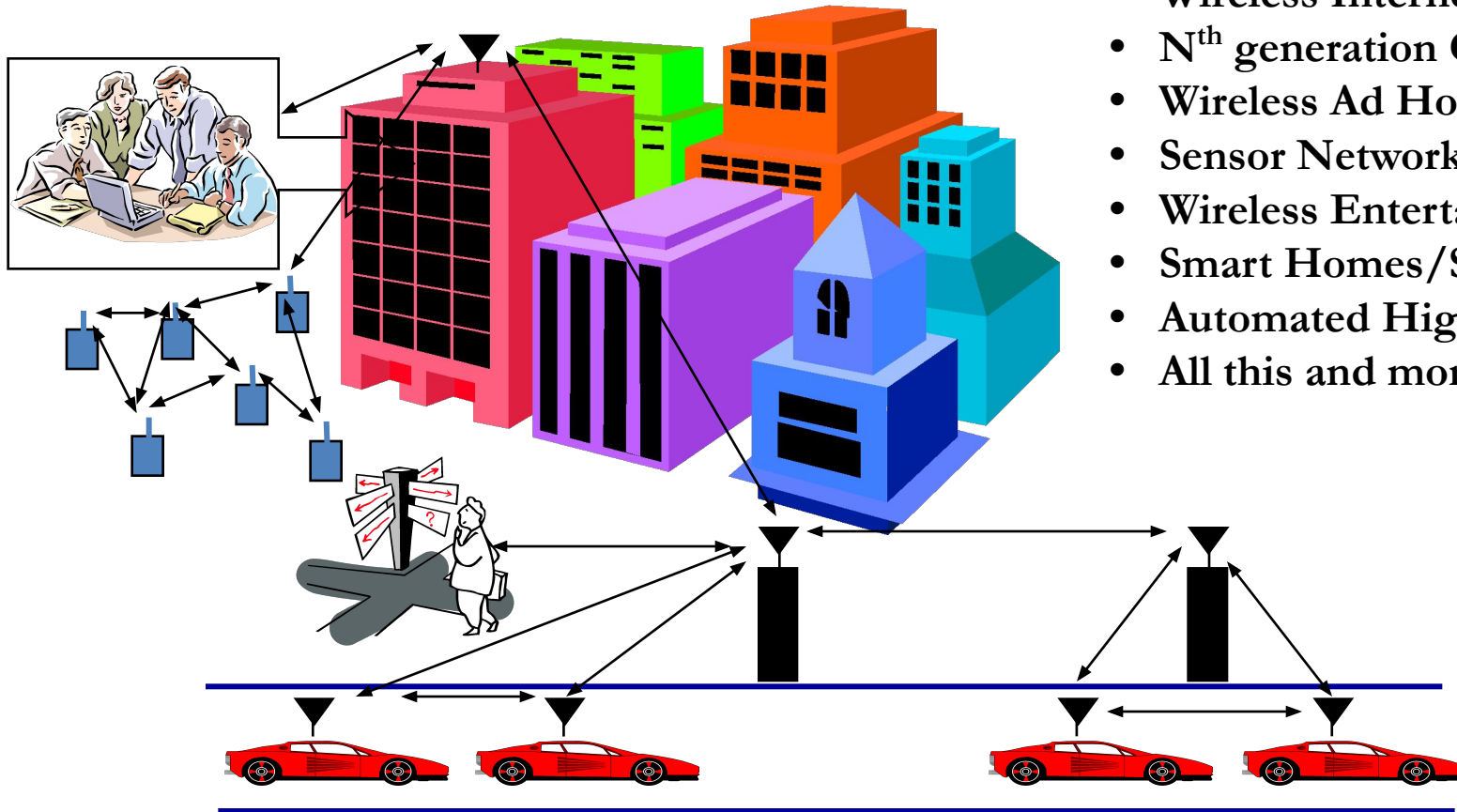
<i>Personal Area Network (PAN)</i>	<i>Local Area Network (LAN)</i>	<i>Metropolitan Area Network (MAN)</i>	<i>Wide Area Network (WAN)</i>
Bluetooth	802.11b 802.11a 802.11g	802.16 802.16a 802.16e	GSM GPRS CDMA 2.5G 3G
Medium data rates 1 Mbps to 2 Mbps	High data rates 11 Mbps to 54 Mbps	Very high data rates Quality of service, up to 280 Mbps	Low to medium data rates 10 Kbps to 2.4 Mbps
Very short range 3 m (~10 feet)	Short range 100 m (~300 feet)	Medium range 50 km (~31 miles)	Long range Global
Notebook to PC to peripherals devices to systems	Computer to computer and the internet	LAN or computer to high speed wire line internet	Smart phones and PDAs to WANs and the internet

Milestones in Wireless Communications



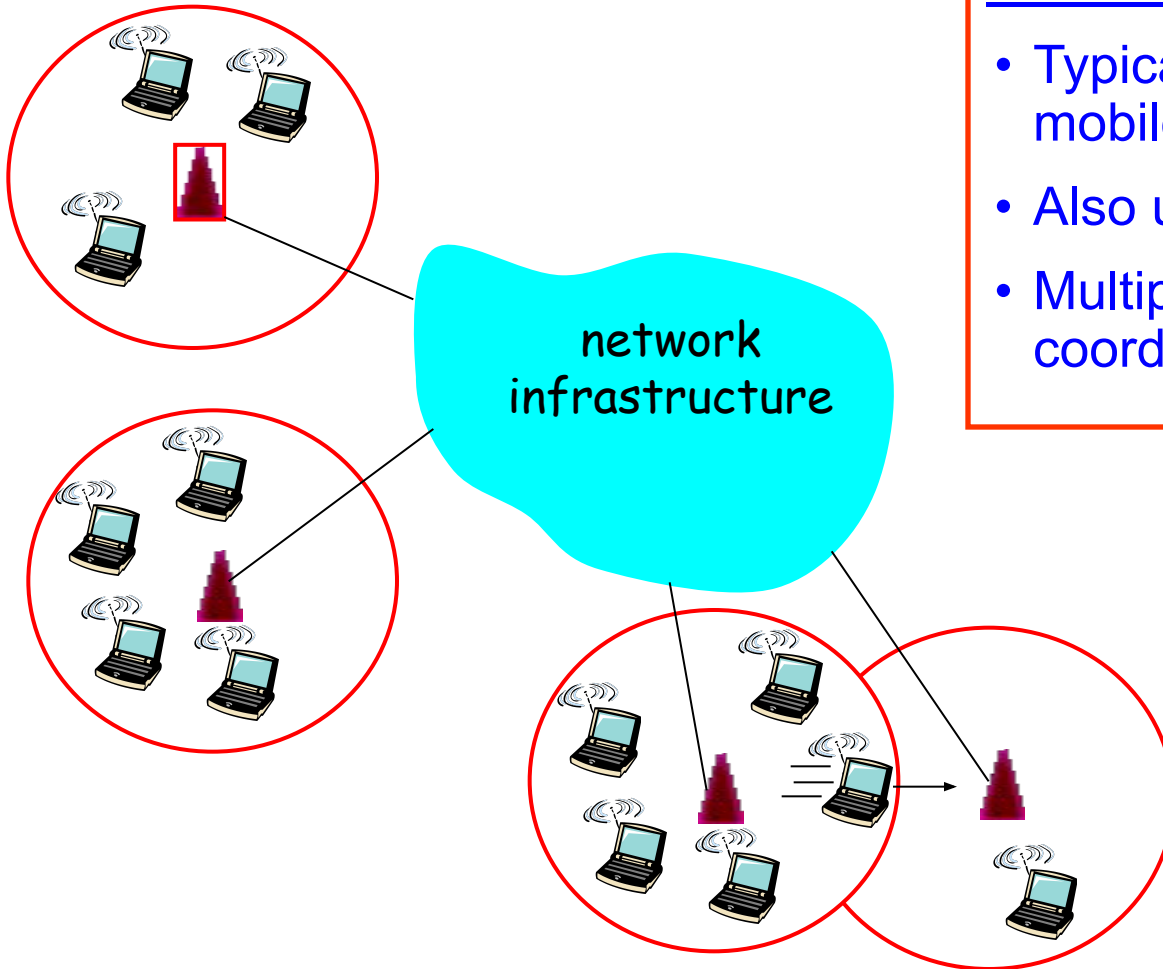
Wireless Networks

Ubiquitous Communication Among People and Devices



- Wireless Internet access
- Nth generation Cellular
- Wireless Ad Hoc Networks
- Sensor Networks
- Wireless Entertainment
- Smart Homes/Spaces
- Automated Highways
- All this and more...

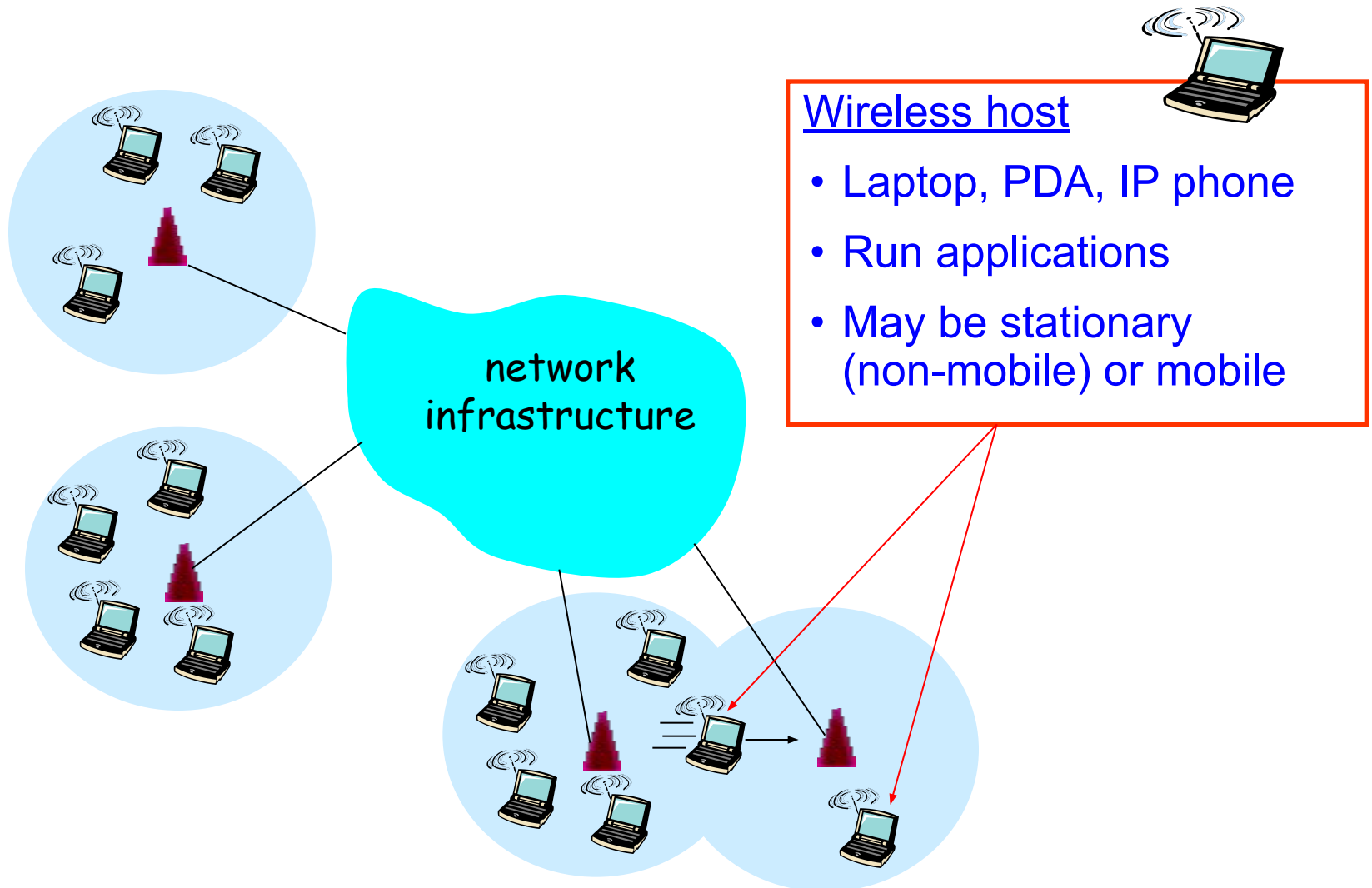
Wireless Network: Wireless Link



Wireless link

- Typically used to connect mobile(s) to base station
- Also used as backbone link
- Multiple access protocol coordinates link access

Wireless Network: Wireless **Hosts**

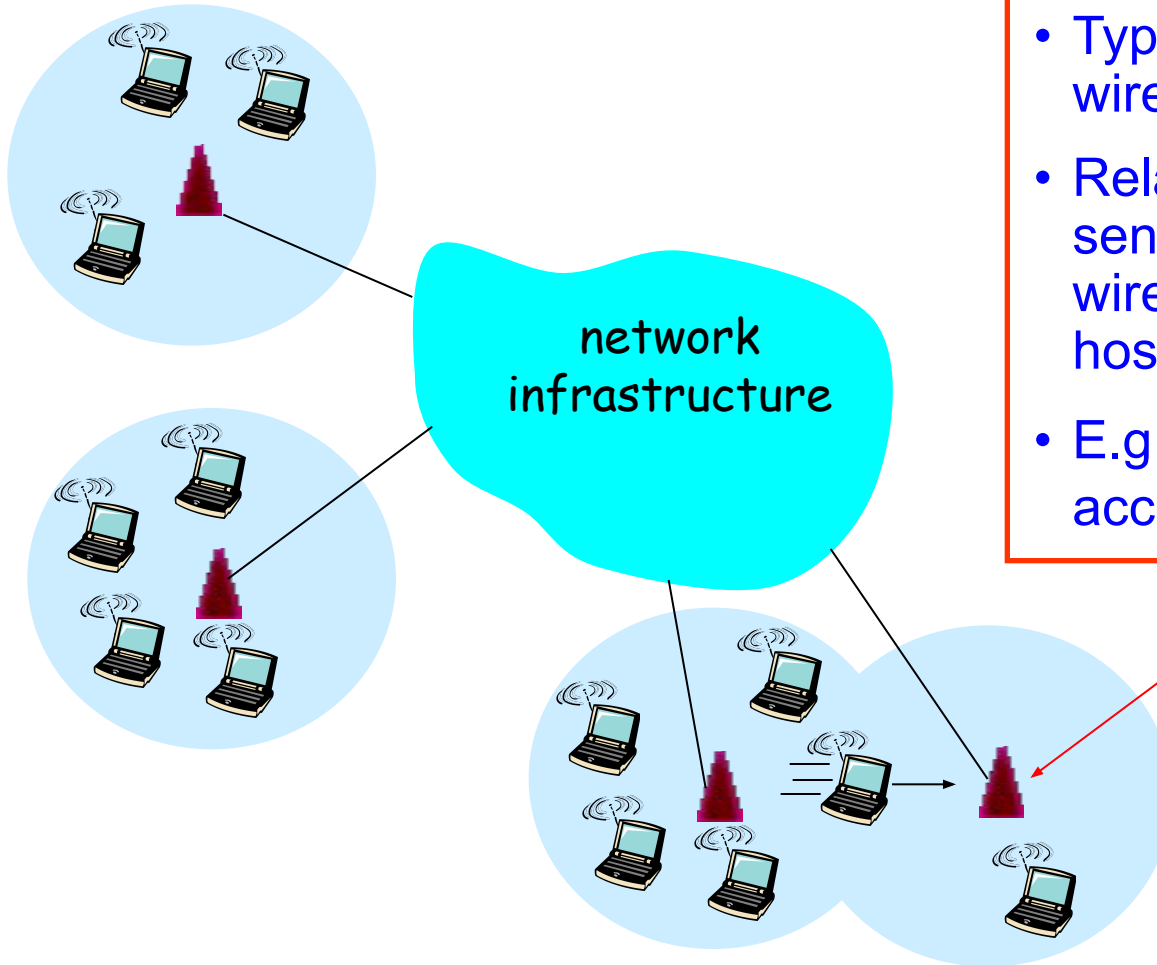


Wireless Network: **Base Station**



Base station

- Typically connected to wired network
- Relay responsible for sending packets between wired network and wireless host(s) in its “area”
- E.g., cell towers, 802.11 access points

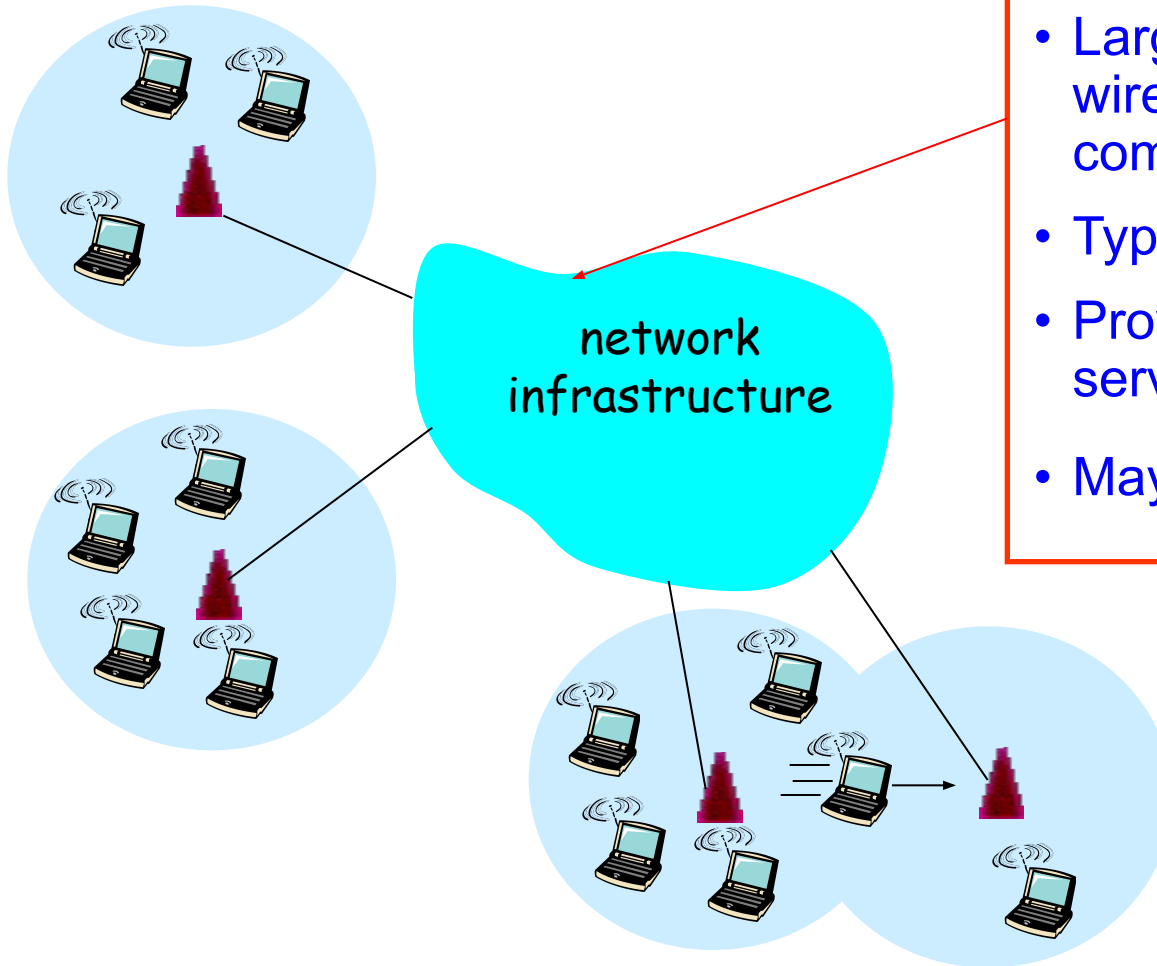


Wireless Network: Infrastructure



Network infrastructure

- Larger network with which a wireless host wants to communicate
- Typically a wired network
- Provides traditional network services
- May not always exist



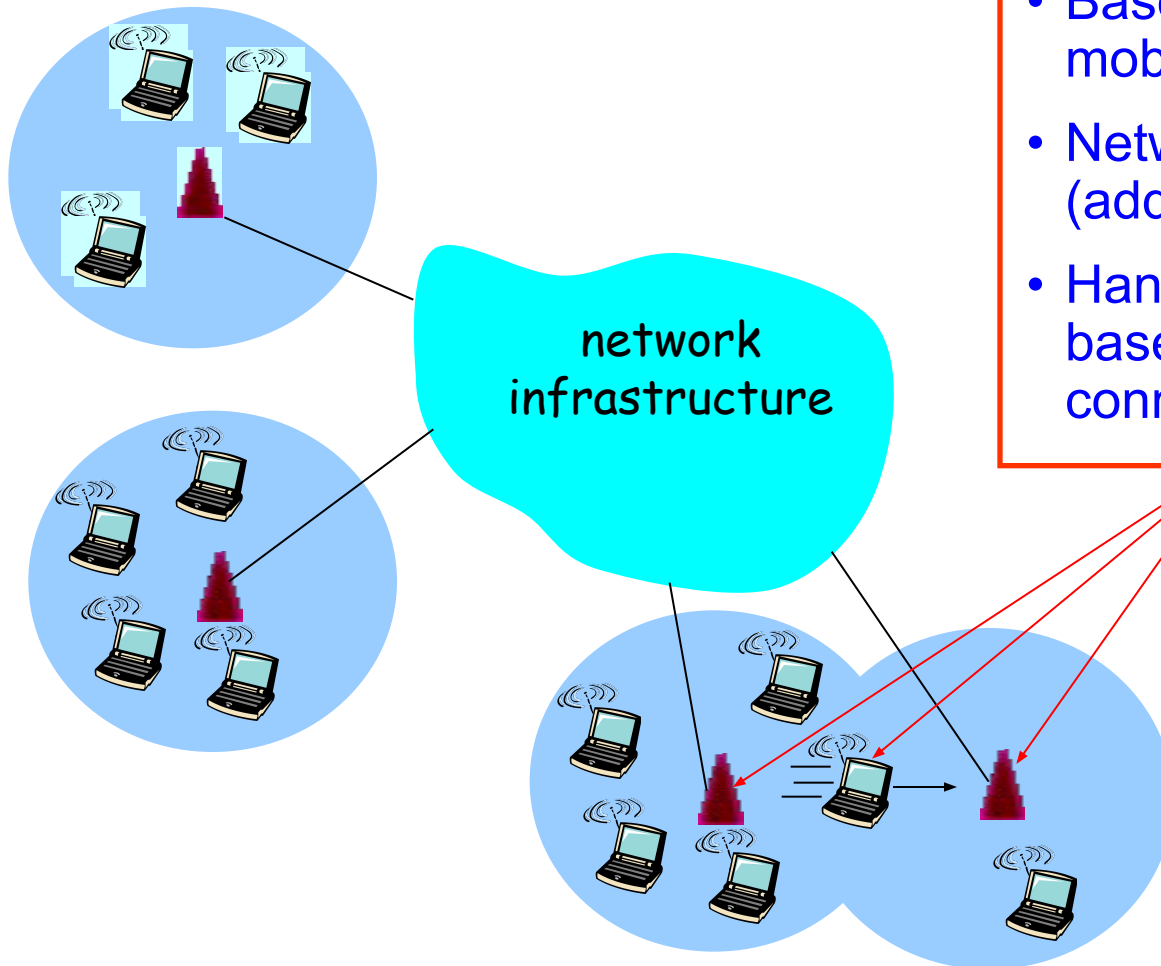
Wireless Access Networks:: Infrastructure Modes

Scenario #1: Infrastructure Mode



Infrastructure mode

- Base station connects mobiles into wired network
- Network provides services (addressing, routing, DNS)
- Handoff: mobile changes base station providing connection to wired network



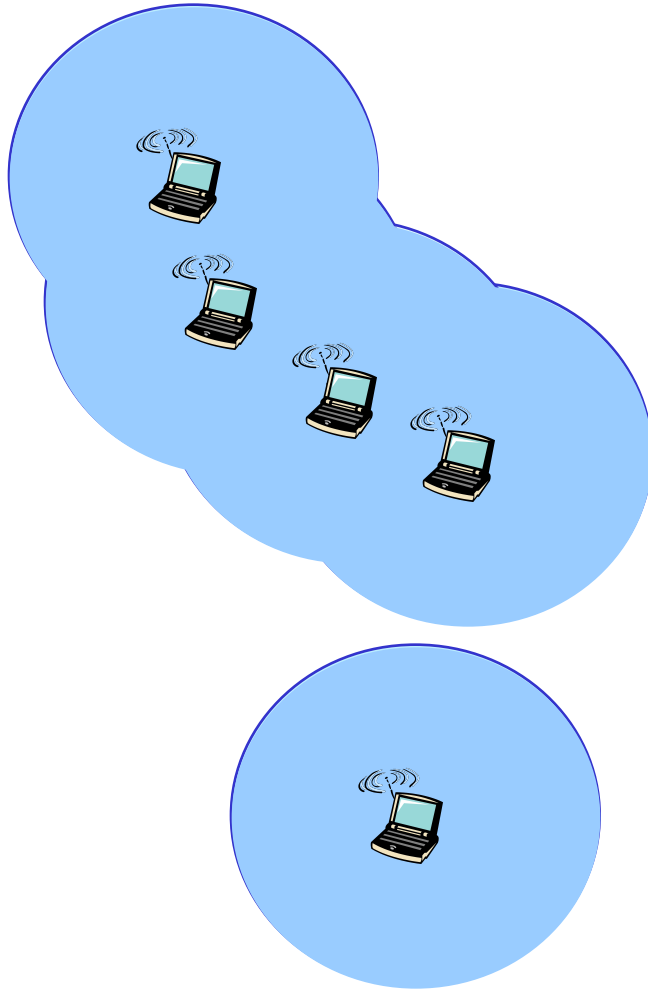
Client-server communication

Wireless Access Networks:: Infrastructure Modes

Scenario #2: Infrastructure-less (Ad Hoc) Mode



Peer-to-peer communication



Ad hoc mode

- No base stations
- Nodes can only transmit to other nodes within link coverage
- Nodes self-organize and route among themselves

Wireless Access Networks:: Infrastructure Modes



	Infrastructure-based
Single-hop	• Base station connected to larger wired network • WiFi based wireless LAN • Cellular telephony networks • WiMAX

Wireless Access Networks:: Infrastructure Modes



	Infrastructure-based	Infrastructure-less
Single-hop	• Base station connected to larger wired network • WiFi based wireless LAN • Cellular telephony networks • WiMAX	• One node coordinates the transmissions of the others • Bluetooth • Ad-hoc 802.11 (e.g., P2P, WiFi Direct)

Wireless Access Networks:: Infrastructure Modes



	Infrastructure-based	Infrastructure-less
Single-hop	• Base station connected to larger wired network • WiFi based wireless LAN • Cellular telephony networks • WiMAX	• One node coordinates the transmissions of the others • Bluetooth • Ad-hoc 802.11 (e.g., P2P, WiFi Direct)
Multi-hop	• Base station exists • Some nodes must relay through other nodes • Wireless Mesh Networks (semi-structured)	

Wireless Access Networks:: Infrastructure Modes



	Infrastructure-based	Infrastructure-less
Single-hop	- Base station connected to larger wired network - WiFi based wireless LAN - Cellular telephony networks - WiMAX	- One node coordinates the transmissions of the others - Bluetooth - Ad-hoc 802.11 (e.g., P2P, WiFi Direct)
Multi-hop	- Base station exists - Some nodes must relay through other nodes - Wireless Mesh Networks (semi-structured)	- No base station exists - Some nodes must relay through others - Ad hoc wireless Networks - Wireless Sensor Networks (Structured / Semi-Structured / Un-Structured :: Depending on Application Nature) - Mobile Ad-hoc Networks / Vehicular Ad-hoc Networks - Delay Tolerant Networks

Wireless Access Networks

Single-hop Infrastructure-based Wireless Systems

1. Cellular Networks
2. Wireless LANs
3. WiMAX

Multi-hop Infrastructure-based Wireless Systems

5. Wireless Mesh Networks (semi-structured)

Single-hop Infrastructure-less Wireless Systems

4. Bluetooth

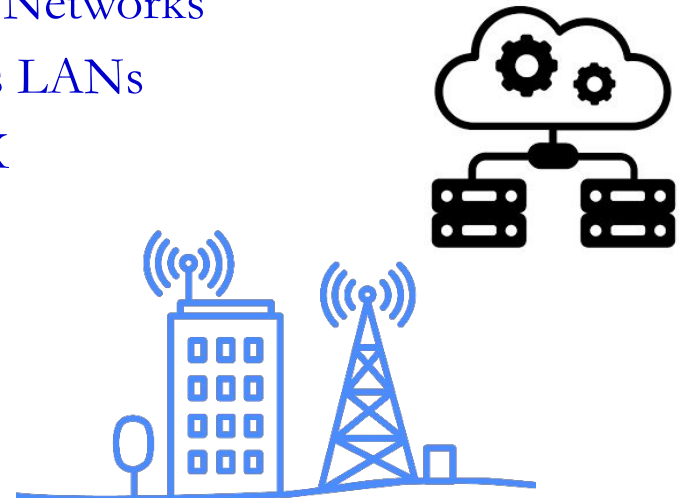
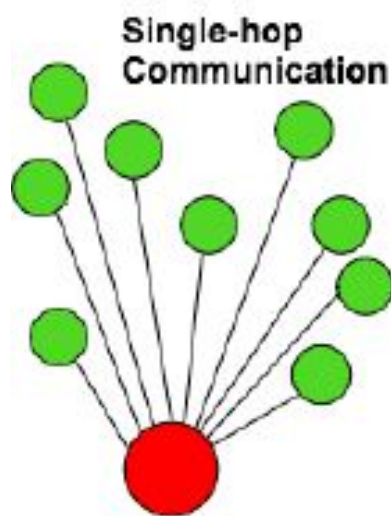
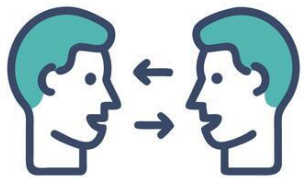
Multi-hop Infrastructure-less Wireless Systems

6. Ad hoc wireless Networks

- 6.1. Wireless Sensor Networks (Structured / Semi-Structured / Un-Structured :: Depending on Application Nature)
- 6.2. Vehicular Ad-hoc Networks
- 6.3. Delay Tolerant Networks

Single-hop Infrastructure-base Wireless Access Networks

1. Cellular Networks
2. Wireless LANs
3. WiMAX



Single-hop Infrastructure-based:: Cellular Networks

- 3G cellular data network connects radio access networks to the public Internet.

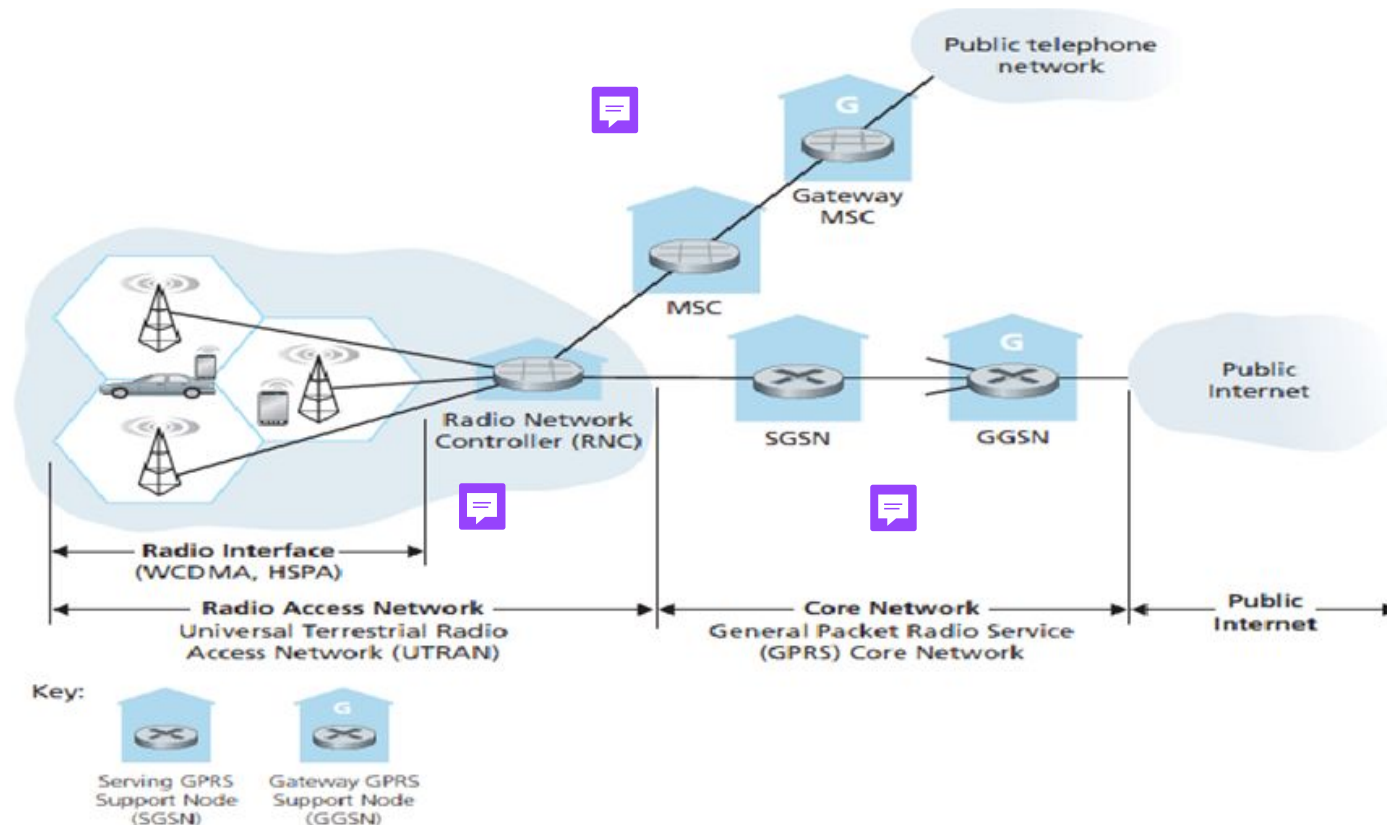
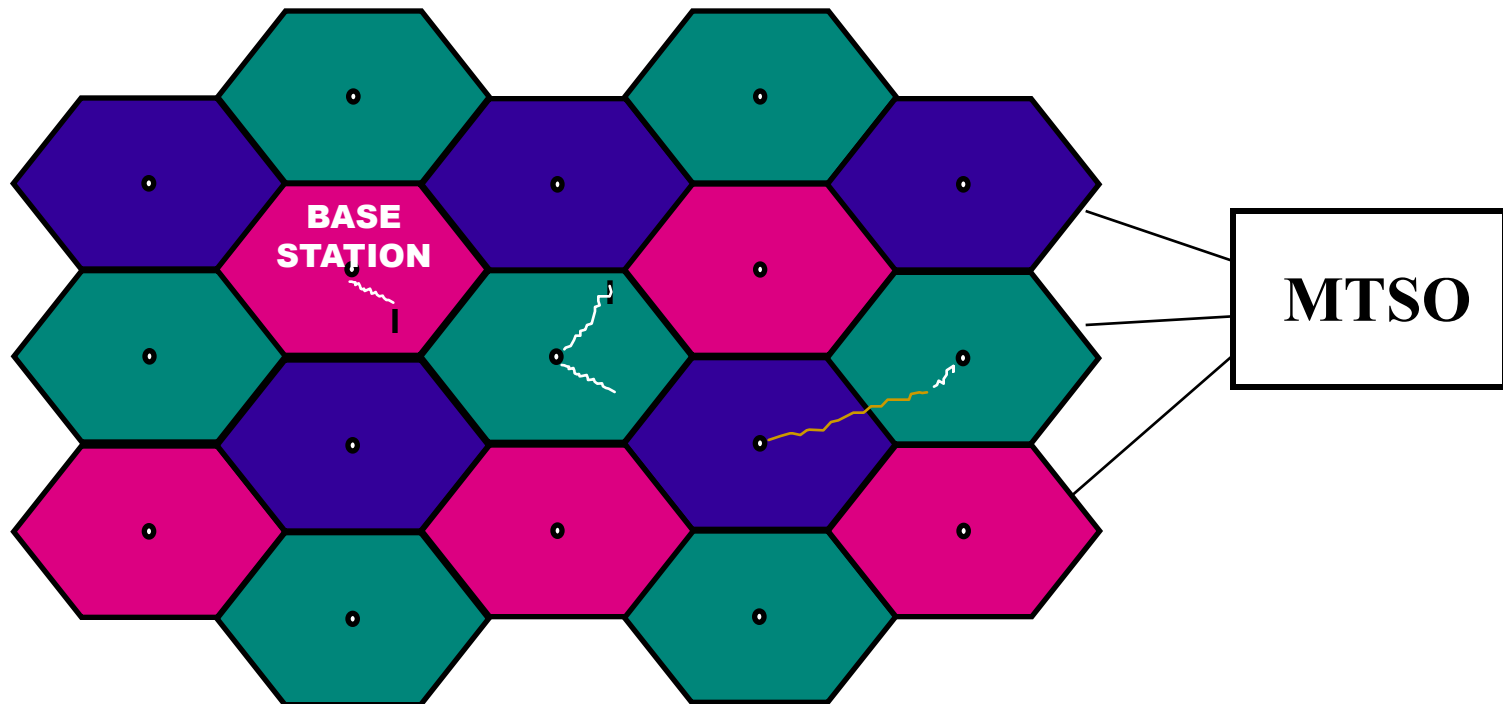


Figure: 3G cellular data network architecture

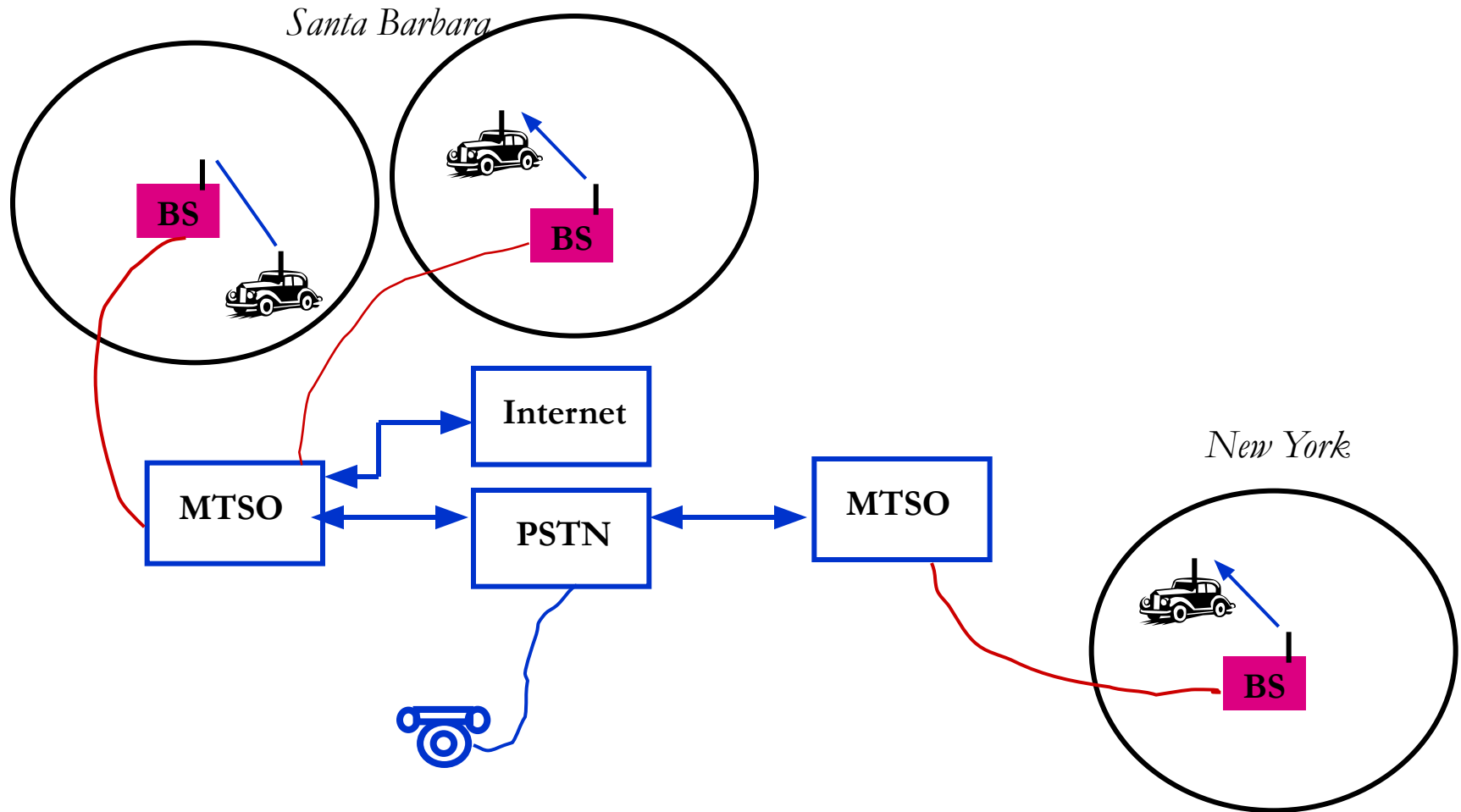
Cellular Networks:

Reuse channels to maximize capacity

- Geographic region divided into cells
- Frequencies/timeslots/codes reused at spatially-separated locations.
- Co-channel interference between same color cells.
- Base stations/MTSOs coordinate handoff and control functions
- Shrinking cell size increases capacity, as well as networking burden



Cellular Networks



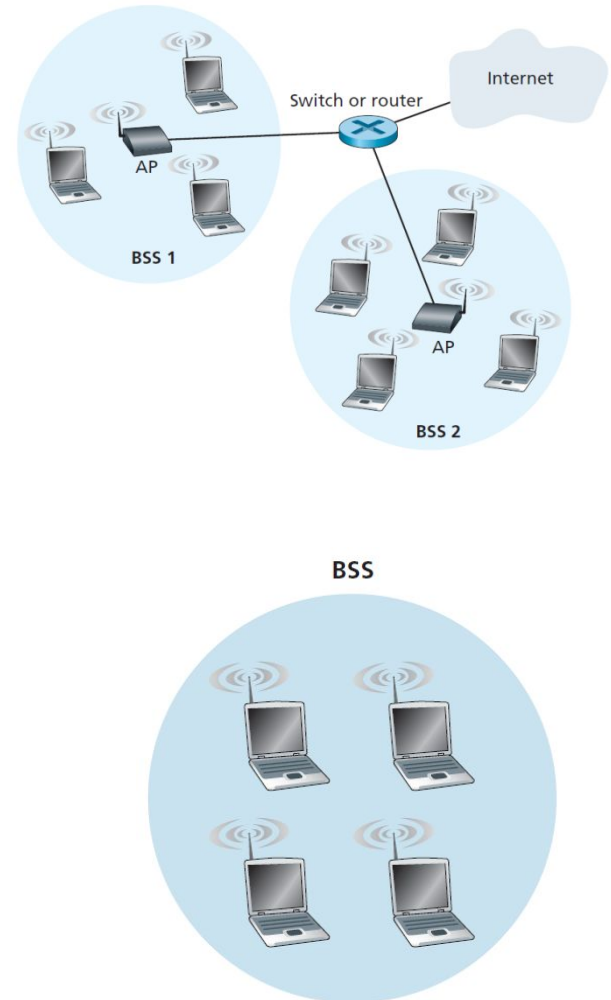
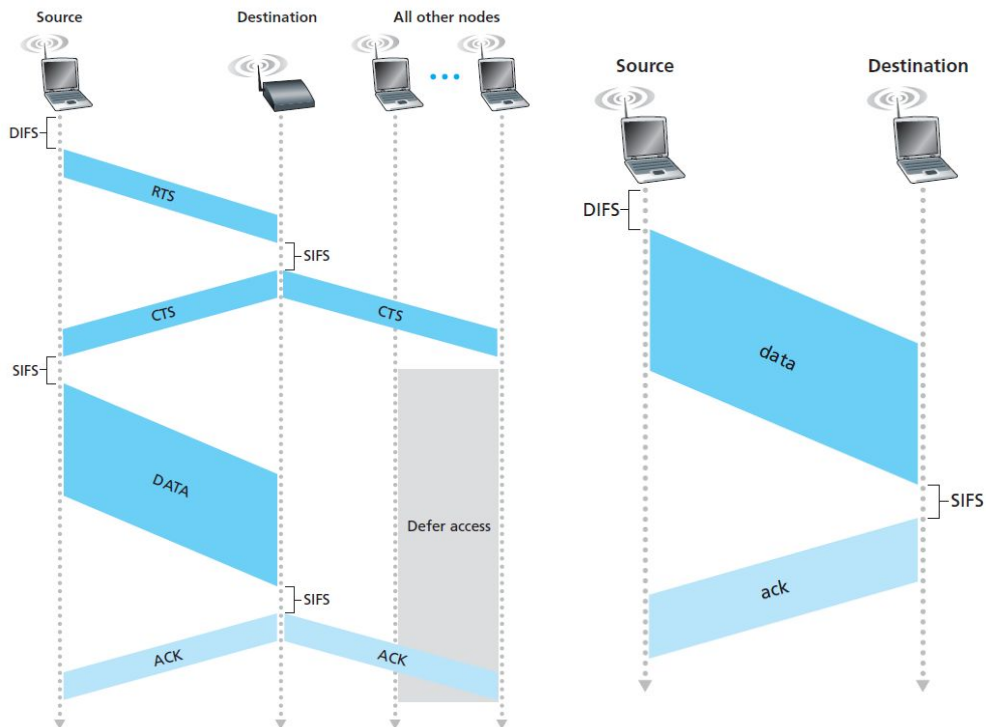
Cellular Design:

Voice and Data

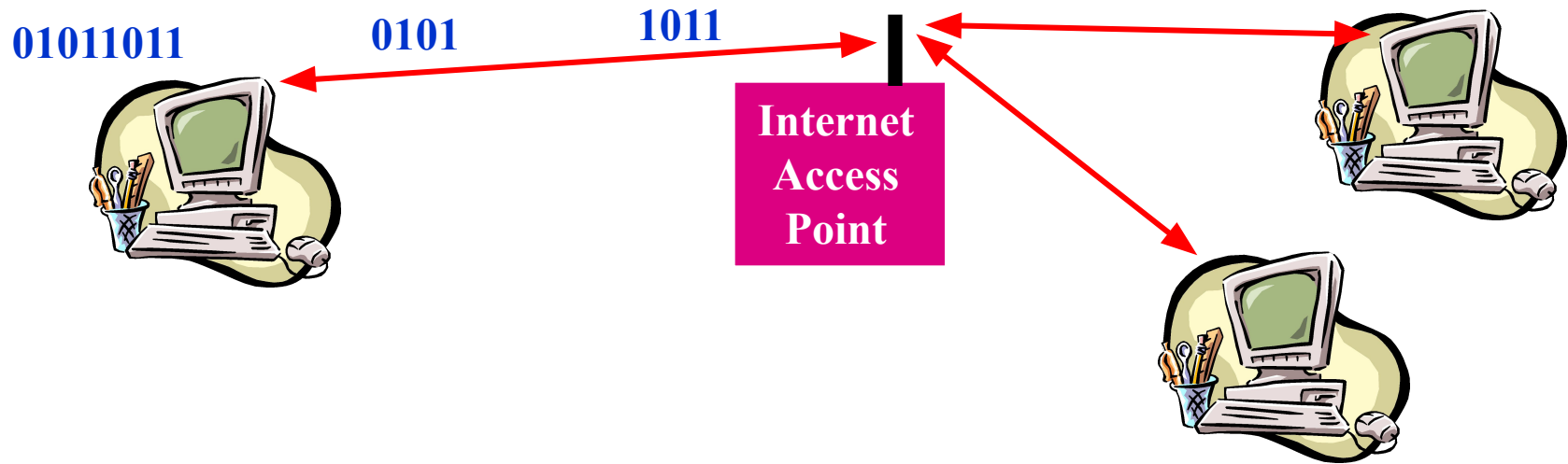
- Data is bursty, whereas voice is continuous
 - Typically require different access and routing strategies
- 3G “widens the data pipe”:
 - Standard based on wideband CDMA
 - Packet-based switching for both voice and data
- Evolution of wireless networks:
 - | GSM
 - | IS-95(CDMA)
 - | CDMA 2000 / W-CDMA // IEEE 802.11a/g/b

Single-hop Infrastructure-based:: Wireless LANs

Standard	Frequency Range	Data Rate
802.11b	2.4–2.485 GHz	up to 11 Mbps
802.11a	5.1–5.8 GHz	up to 54 Mbps
802.11g	2.4–2.485 GHz	up to 54 Mbps



Single-hop Infrastructure-based: Wireless LANs



- WLANs connect “local” computers
- Channel access is shared (random access)
- Backbone Internet provides best-effort service

Single-hop Infrastructure-based:: Wireless LAN Standards

- 802.11b
 - Standard for 2.4GHz ISM band (80 MHz)
 - Frequency hopped spread spectrum
 - 1.6-10 Mbps, 500 ft range
- 802.11a
 - Standard for 5GHz NII band (300 MHz)
 - OFDM with time division
 - 20-70 Mbps, variable range
 - Similar to HiperLAN in Europe
- 802.11g
 - Standard in 2.4 GHz and 5 GHz bands
 - OFDM
 - Speeds up to 54 Mbps

Single-hop Infrastructure-based:: Wireless LAN Standards

Data rate :

- IEEE 802.11b: 11 Mbps
- IEEE 802.11g: 54 Mbps
- IEEE 802.11n (Wi-Fi 4): Up to 600 Mbps
- IEEE 802.11ac (Wi-Fi 5): Up to 6.9 Gbps
- IEEE 802.11ax (Wi-Fi 6/6E): Up to 9.6 Gbps
- IEEE 802.11be (Wi-Fi 7) (emerging): Targeting >30 Gbps

Single-hop Infrastructure-based::

WiMAX

(Worldwide Interoperability for Microwave Access)

- A wireless communications standard - designed to provide 30 to 40 megabit-per-second data rates, with the 2011 update providing up to 1 Gbit/s for fixed stations.
- It refers to interoperable implementations of the IEEE 802.16 family of wireless-networks standards.

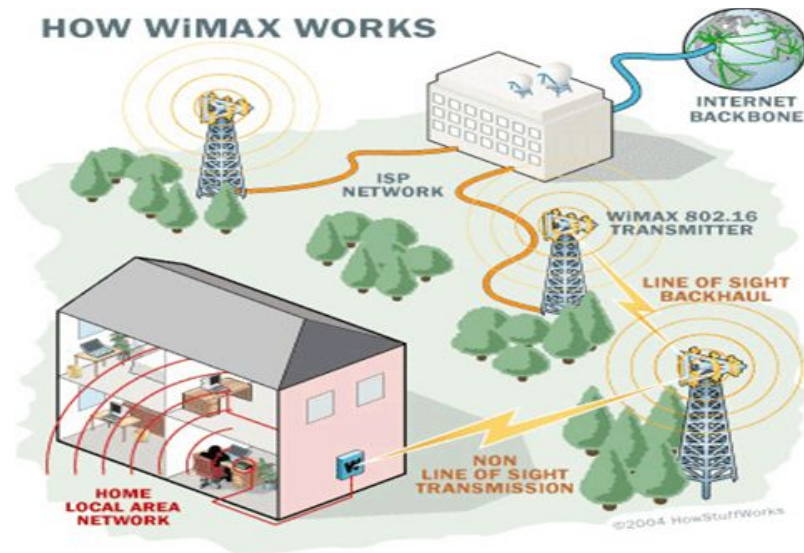


Figure: Working Procedure of WiMAX

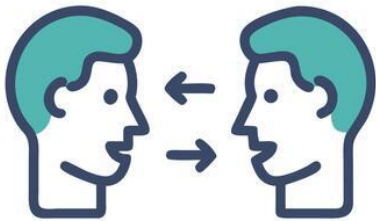
Single-hop Infrastructure-based::

WiMAX

- Provide portable mobile broadband connectivity across cities and countries
- Wireless alternative to cable and digital subscriber line (DSL) for "last mile" broadband access
- Provide data, telecommunications (VoIP) and IPTV services (triple play)
- A source of Internet connectivity as part of a business continuity plan

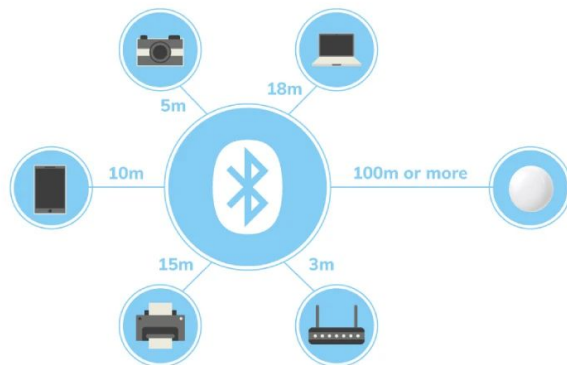
Single-hop Infrastructure-less Wireless Access Networks

4. Bluetooth

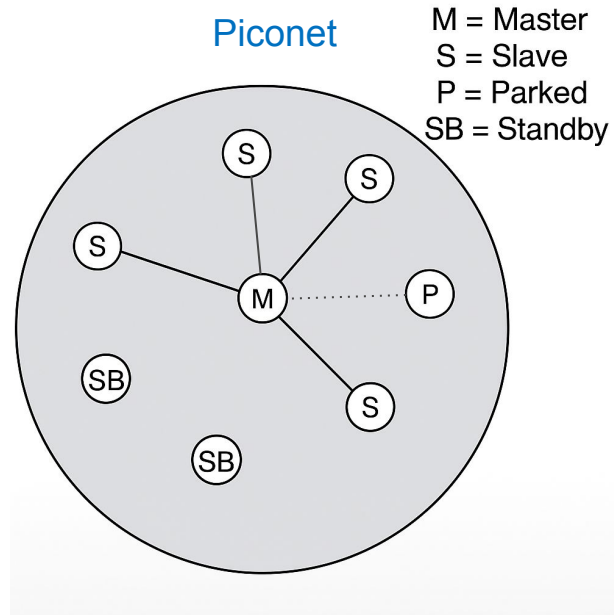


Single-hop Infrastructure-less::

Bluetooth Networks



Piconet



Master – Slave Communication

only the Master can initiate and control communication within a piconet.

Parked Node: Devices are still synchronized with the master but are **not actively participating** in communication.

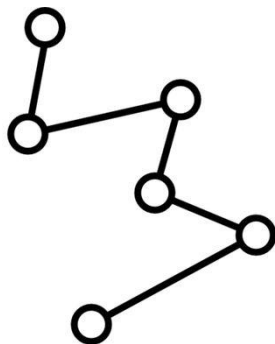
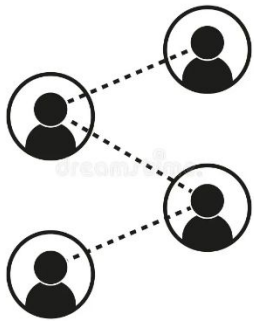
Standby Node: Devices are in an **idle state**, **not connected** to any piconet.

Single-hop Infrastructure-less: Bluetooth Networks

Property	Description
Frequency Band	2.4 GHz ISM band (2.402 – 2.480 GHz)
Technology	Frequency Hopping Spread Spectrum (FHSS), 1600 hops/sec
Network Topology	Piconet (1 master + up to 7 active slaves)
Data Rate	Up to 721 Kbps (Classic)
Range	~10 meters (Class 2), up to 100 meters (Class 1)
Roles	Master (initiates & controls communication), Slave (responds to master)
Security	Authentication, Authorization, Encryption
Voice Support	Up to 3 simultaneous SCO voice channels
Device States	Active, Parked, Standby

Multi-hop Infrastructure-base Wireless Access Networks

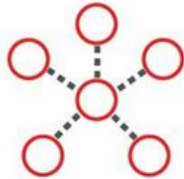
5. Wireless Mesh Networks (semi-structured)



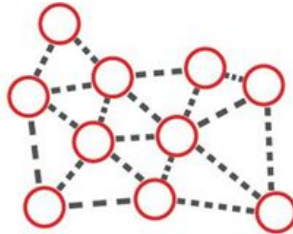
Multi-hop Infrastructure-based:: Wireless Mesh Networks



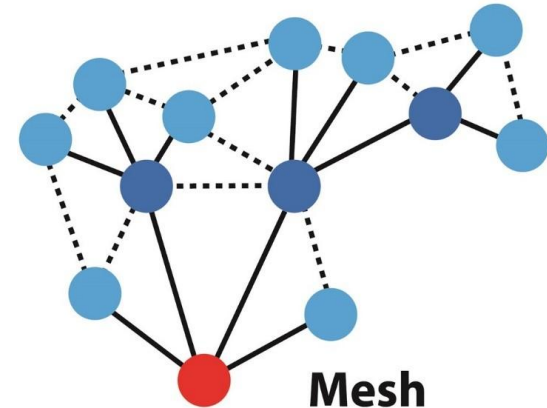
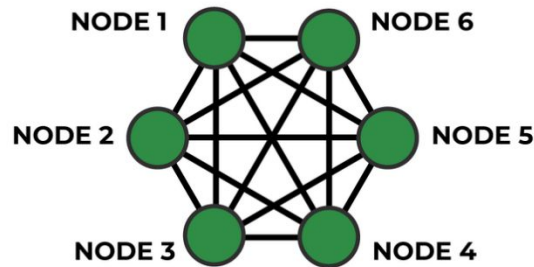
Point to point



Star

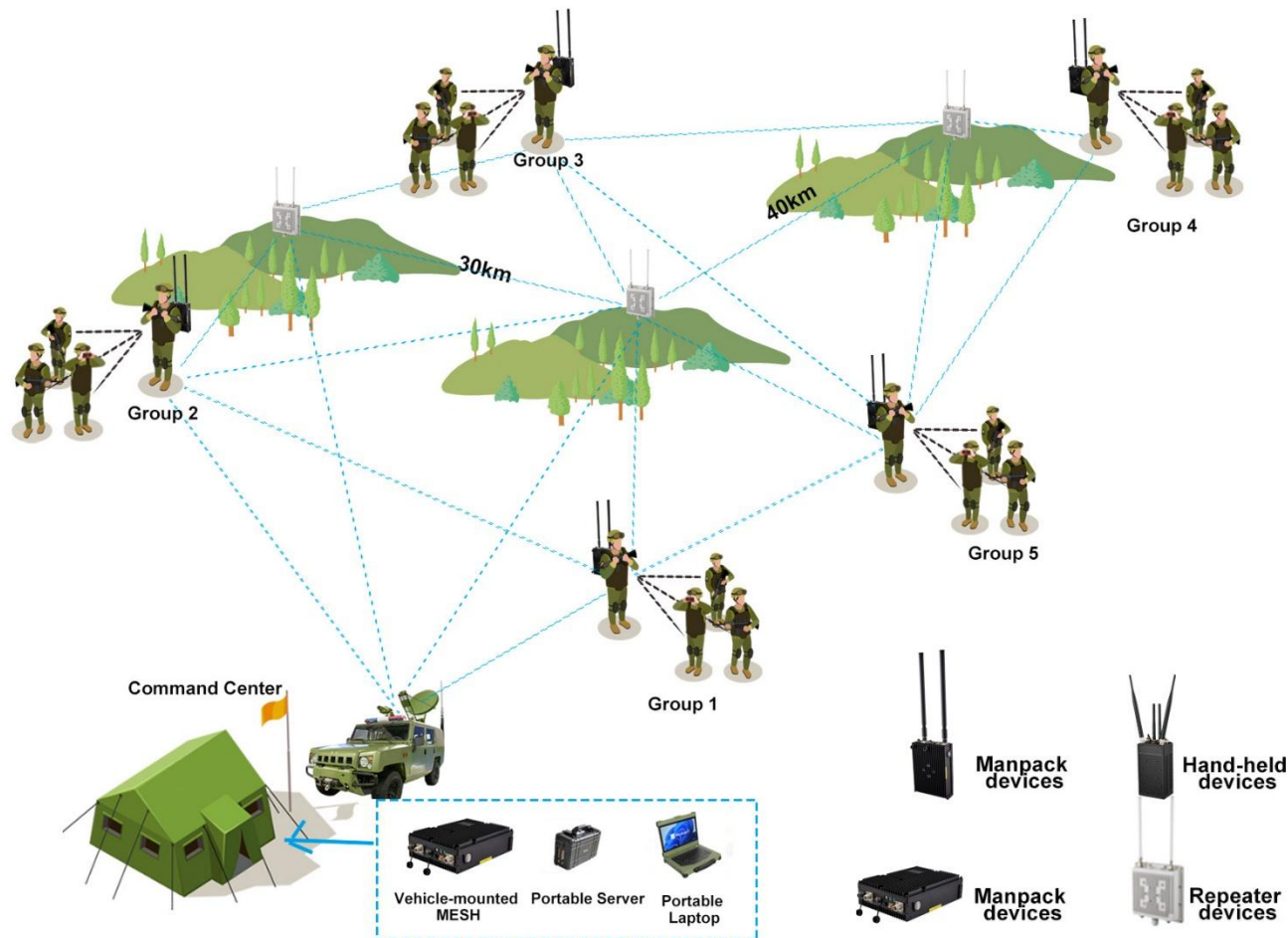


Mesh



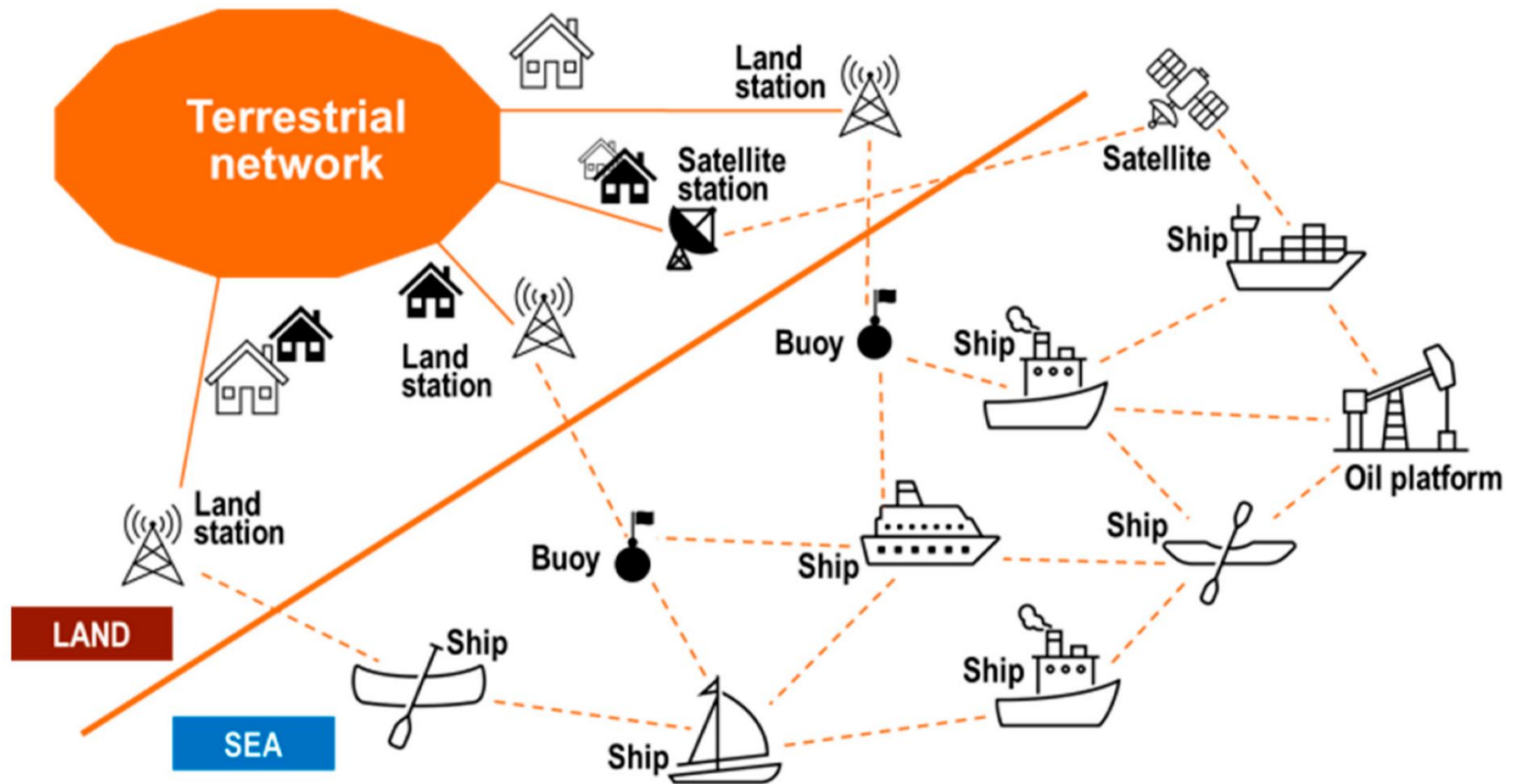
Multi-hop Infrastructure-based:: Wireless Mesh Networks

Application: Military battlefield communication



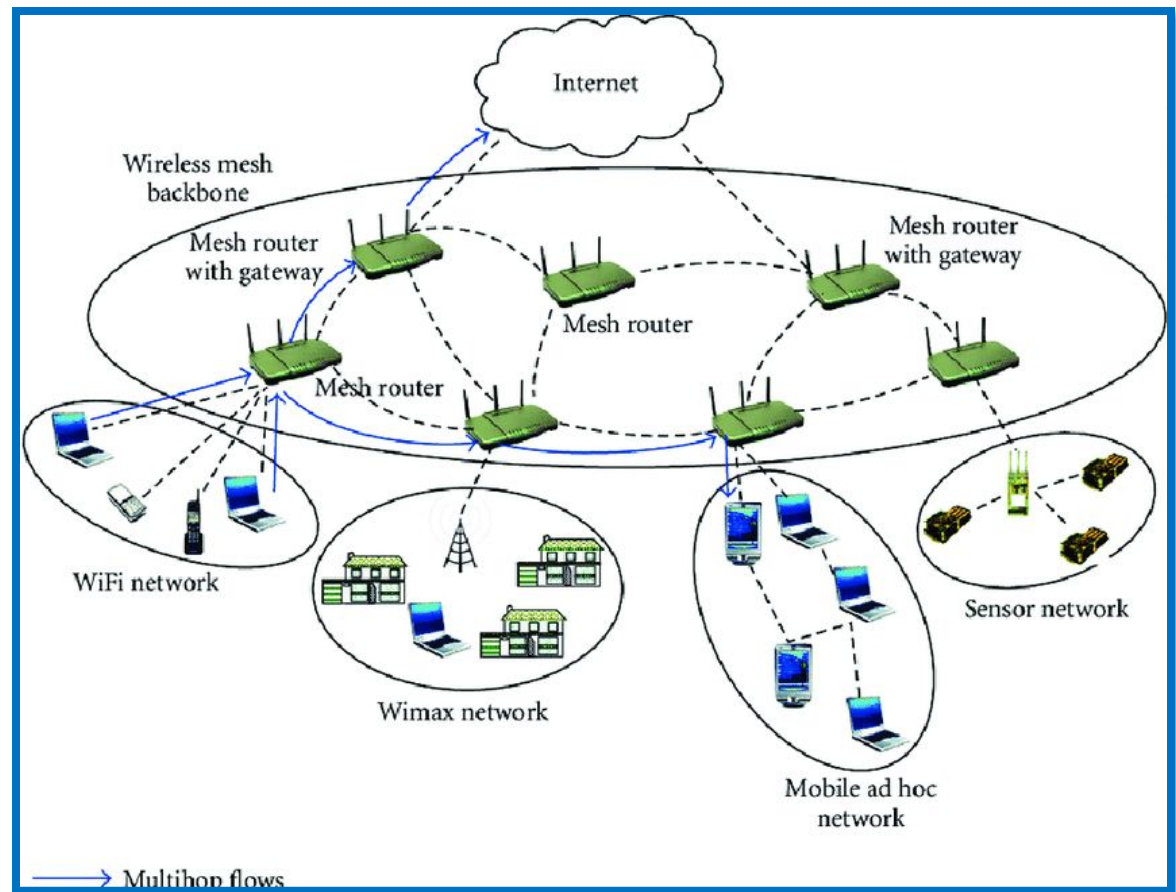
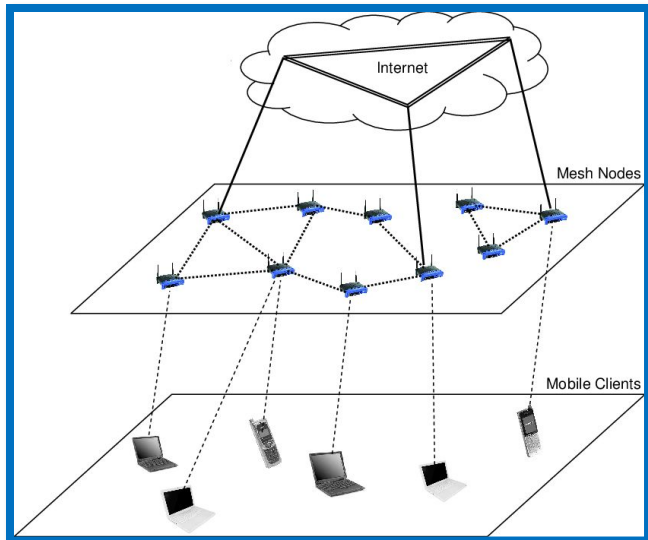
Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Maritime Mesh Network



Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Hybrid Wireless Mesh Network for Backbone for Community Broadband Connectivity



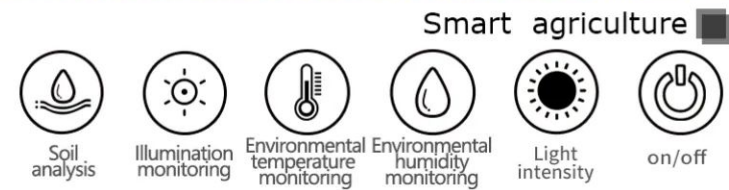
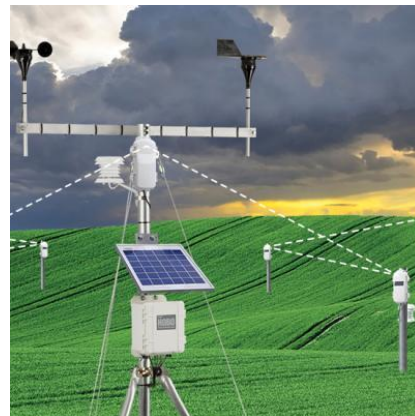
Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Drone Mesh Networks (UAV Mesh Networks)



Multi-hop Infrastructure-based:: Wireless Mesh Networks

Application: **Very Large-Scale** Agricultural Monitoring with **Mobile** Machineries



Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Smart & Connected City with Mobile & Static Devices



Multi-hop Infrastructure-based:: Wireless Mesh Networks

How WMNs Work

1. Network Structure

WMNs consist of two types of devices:

- **Mesh Routers (MRs):**

- Fixed nodes that form the backbone of the mesh.
- Often connected to the internet.

- **Mesh Clients (MCs):**

- End-user devices like laptops, phones, sensors.
- Connect to the mesh routers or other clients.

2. Multi-hop Routing

- Data travels **hop-by-hop** through multiple mesh routers to reach its destination.
- If a direct path is not available, packets are forwarded through **intermediate nodes**.

3. Self-Healing Capability

- If a node fails or a link breaks, the network **automatically reroutes** traffic through alternative paths.
- Ensures **high reliability and robustness**.

4. Self-Configuration

- New nodes can join the network and start routing **without manual configuration**.
- Each node participates in the routing process and **discovers neighbors dynamically**.

Multi-hop Infrastructure-based::

Wireless Mesh Networks



Key Features:

Feature	Description
Multi-hop	Increases coverage without requiring all nodes to be in range of each other
Self-healing	Network adapts to failures automatically
Flexible coverage	Nodes can be added easily to expand range
No central controller	Decentralized, peer-based communication

Multi-hop Infrastructure-based: Wireless Mesh Networks

Key Challenges:

1. Interference and Shared Medium

Problem: Nodes use a shared wireless medium, leading to co-channel interference, hidden terminal, and exposed terminal problems.

Impact: Reduces throughput and increases collisions.

2. Scalability

Problem: As the number of nodes increases, routing overhead and contention grow.

Impact: Performance degrades due to increased latency and reduced bandwidth per node.

3. Dynamic Topology & Routing Complexity

Problem: Nodes may join, leave, or move, causing frequent route breaks and topology changes.

Impact: Requires adaptive routing protocols and fast convergence.

4. Limited Bandwidth and Throughput

Problem: Each additional hop in a path reduces available bandwidth due to medium reuse.

Impact: End-to-end throughput suffers, especially in large mesh networks.

Multi-hop Infrastructure-based:: Wireless Mesh Networks

Key Challenges:

5. Security Threats

Problem: WMNs are vulnerable to eavesdropping, spoofing, DoS attacks, and routing attacks.

Impact: Compromises confidentiality, integrity, and availability.

6. Quality of Service (QoS)

Problem: Hard to guarantee latency, jitter, and packet delivery ratio over unreliable, shared wireless links.

Impact: Affects performance of real-time applications like VoIP and video streaming.

7. Synchronization and Coordination

Problem: Distributed nature of WMNs makes scheduling transmissions and synchronizing clocks challenging.

Impact: Causes collisions, inefficient channel use, and poor coordination.

8. Heterogeneity of Devices

Problem: WMNs may include devices with different capabilities, interfaces, and protocols.

Impact: Interoperability issues arise, complicating routing and communication.

Multi-hop Infrastructure-based:: Wireless Mesh Networks

Watch: [How do Mesh Networks Works?](#)

Watch: [Mesh Networking Idea](#)

Watch: [Mesh WiFi Vs. WiFi Extenders](#)

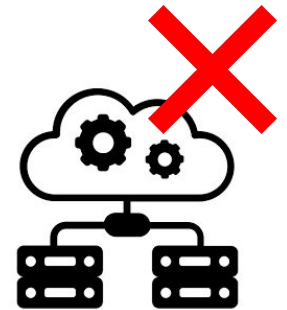
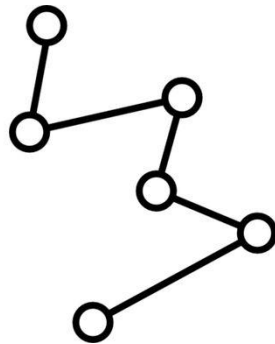
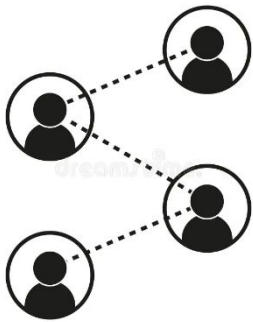
Multi-hop Infrastructure-less Wireless Access Networks

6. Ad hoc wireless Networks

6.1. Wireless Sensor Networks

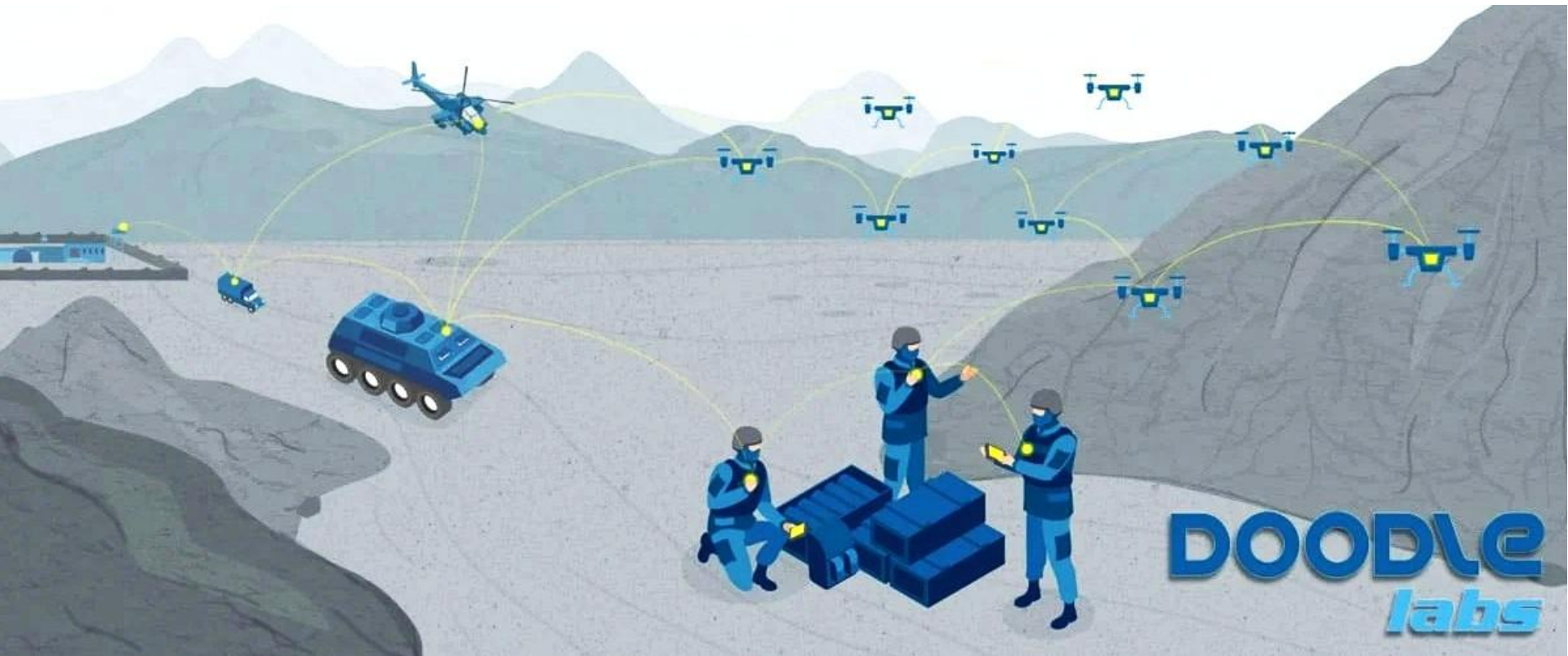
6.2. Mobile Ad-hoc Networks / Vehicular Ad-hoc Networks

6.3. Delay Tolerant Networks



Multi-hop Infrastructure-less:: Wireless Ad-Hoc Networks

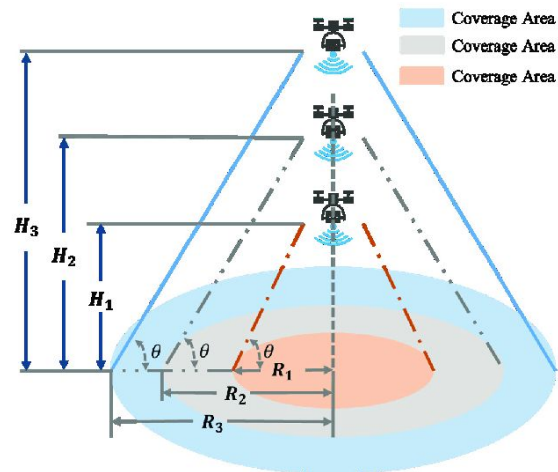
Application: Military battlefield communication in Ad-Hoc Mode



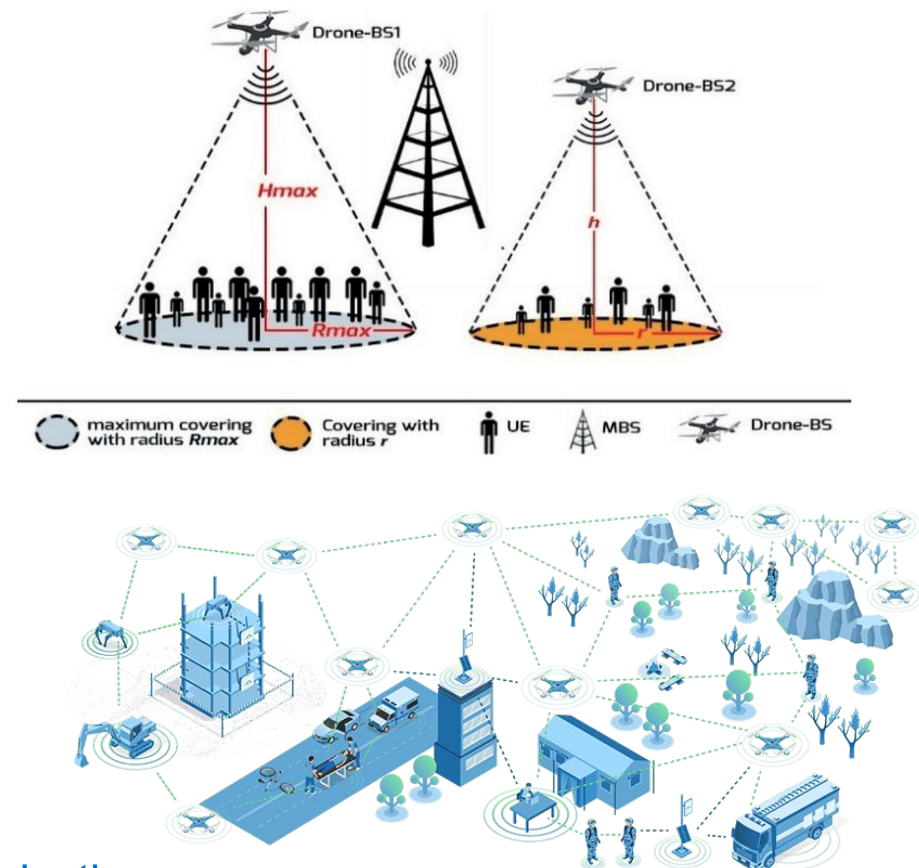
Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

Application: Drone Ad-Hoc Networks (UAV Ad-Hoc Networks)

- Temporary Internet Coverage

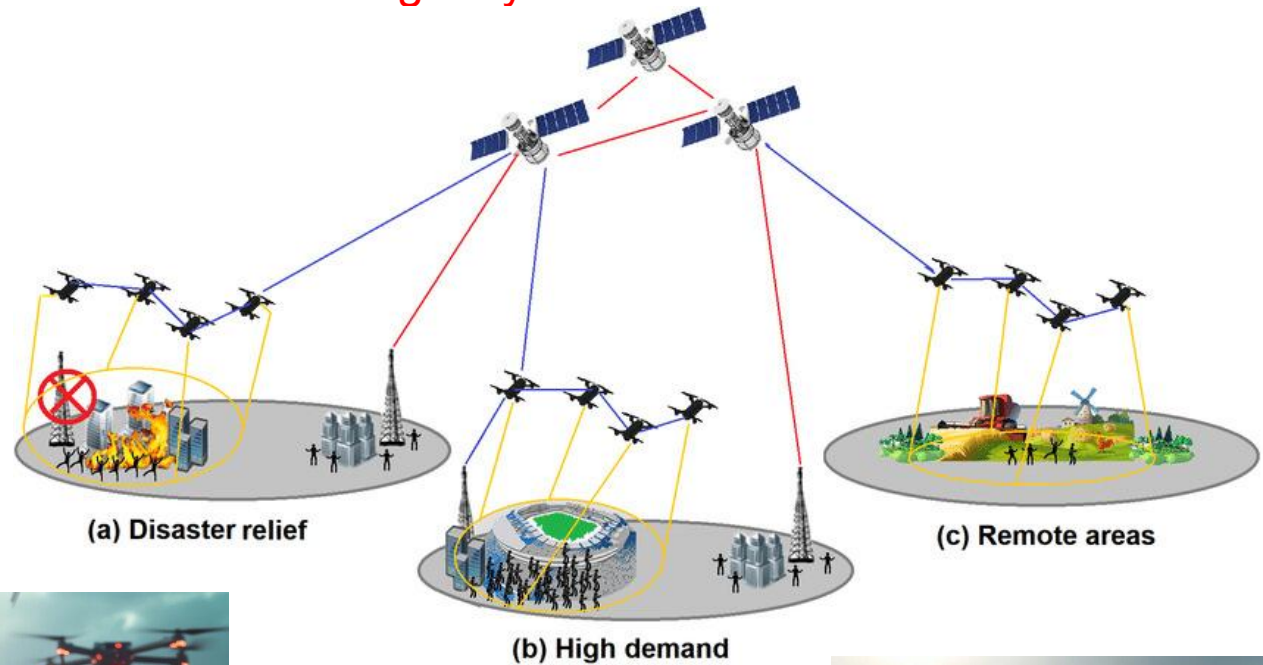


- Search and Rescue in Remote Areas
- Environmental and Wildlife Monitoring
- Disaster Response & Emergency Communications



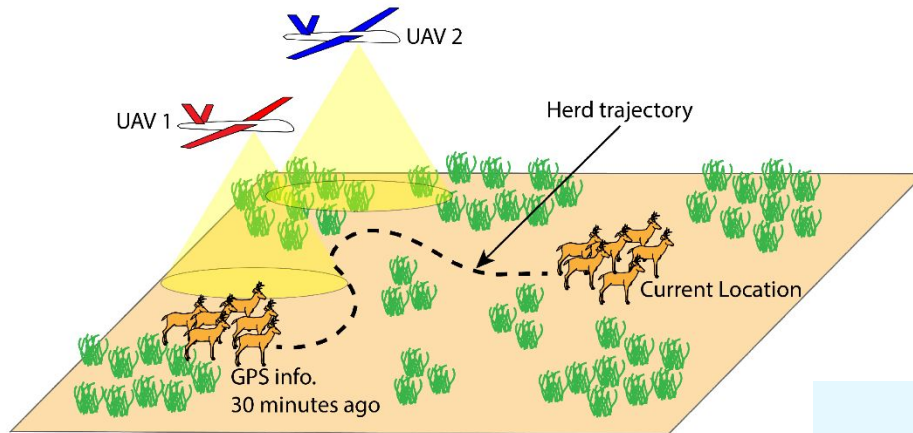
Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

Application: Search and Rescue & Emergency Communications in Remote Areas

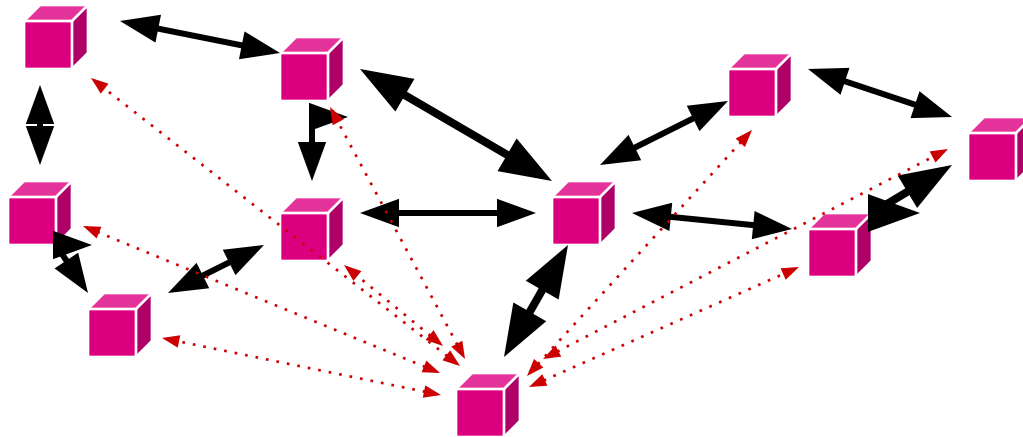


Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

Application: **Environmental and Wildlife Monitoring**



Multi-hop Infrastructure-less: Ad-Hoc Networks



- Peer-to-peer communications.
- No backbone infrastructure.
- Routing can be multi-hop.
- Topology is dynamic.
- Fully connected with different link SINRs

Multi-hop Infrastructure-less::

Ad-Hoc Networks

- Ad-hoc networks provide a flexible network infrastructure for many emerging applications.
- The capacity of such networks is generally unknown.
- Transmission, access, and routing strategies for ad-hoc networks are generally ad-hoc.
- Cross layer design critical are very challenging.
- Energy constraints impose interesting design tradeoffs for communication and networking.

Multi-hop Infrastructure-less: Wireless Mesh Network vs Wireless Ad-Hoc Networks

Aspect	Wireless Mesh Network (WMN)	Wireless Ad-hoc Network (WAHN)
Infrastructure	Semi-infrastructure (uses fixed mesh routers as backbone)	Purely infrastructure-less
Node Roles	Distinct roles: Mesh routers (static) and clients (dynamic)	All nodes are equal; any node can route data
Deployment	Pre-planned, often permanent	Temporary, spontaneous setup
Routing	Stable and optimized (e.g., HWMP, OLSR)	Dynamic and reactive (e.g., AODV, DSR)
Scalability	Highly scalable due to backbone support	Limited scalability due to overhead and congestion
Mobility	Clients may be mobile; routers are often fixed	All nodes may be mobile
Reliability	More reliable due to fixed routers and redundant paths	Less reliable, frequent topology changes
Use Cases	Community networks, smart cities, campus-wide networks	Military ops, disaster relief, emergency response

Multi-hop Infrastructure-less: Wireless Mesh Network vs Wireless Ad-Hoc Networks

Aspect	Wireless Mesh Network (WMN)	Wireless Ad-hoc Network (WAHN)
Radio Interfaces	Often multi-radio and multi-channel (e.g., 802.11s, WiMAX, LTE)	Usually single-radio, single-channel (e.g., 802.11, Bluetooth)
Backhaul Support	Yes, mesh routers may connect to the Internet or backbone	No backhaul, only peer-to-peer communication
Gateway Capability	Mesh routers often serve as gateways to external networks	No defined gateway; all nodes are peers
Energy Source	Often powered (e.g., fixed routers)	Often battery-operated (mobile nodes)

Multi-hop Infrastructure-less: Wireless Mesh Network vs Wireless Ad-Hoc Networks

Aspect	Wireless Mesh Network (WMN)	Wireless Ad-hoc Network (WAHN)
Primary Goal	Provide last-mile connectivity and reliable internet access over large areas	Enable peer-to-peer communication in infrastructure-less and mobile environments
Design Focus	Stability, scalability, and long-term connectivity	Quick deployment, flexibility, and dynamic adaptability
Application Focus	Community WiFi, smart cities, industrial IoT, public safety	Military operations, disaster relief, vehicular networks
Traffic Pattern	Typically many-to-one or client-gateway	Dynamic patterns; often all-to-all or many-to-many
Performance Goal	High throughput, fault tolerance, consistent QoS	Fast topology adaptation, minimal delay in dynamic environments

Multi-hop Infrastructure-less:: Wireless Ad-Hoc Networks

6. Ad hoc wireless Networks

6.1. Wireless Sensor Networks

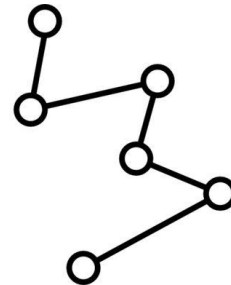
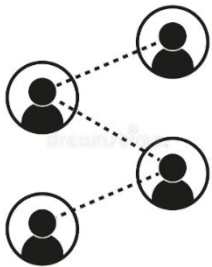
6.2. Mobile Ad-hoc Networks /Vehicular Ad-hoc Networks

6.3. Delay Tolerant Networks

Feature	WSN	VANET	DTN
Infrastructure-less (or minimal)	✓	✓	✓
Multi-hop communication	✓	✓	✓
Dynamic topology	✓ (due to node failure/sleep)	✓ (due to vehicle mobility)	✓ (due to intermittent connectivity)
Self-organizing	✓	✓	✓
Decentralized control	✓	✓	✓

Wireless Sensor Networks

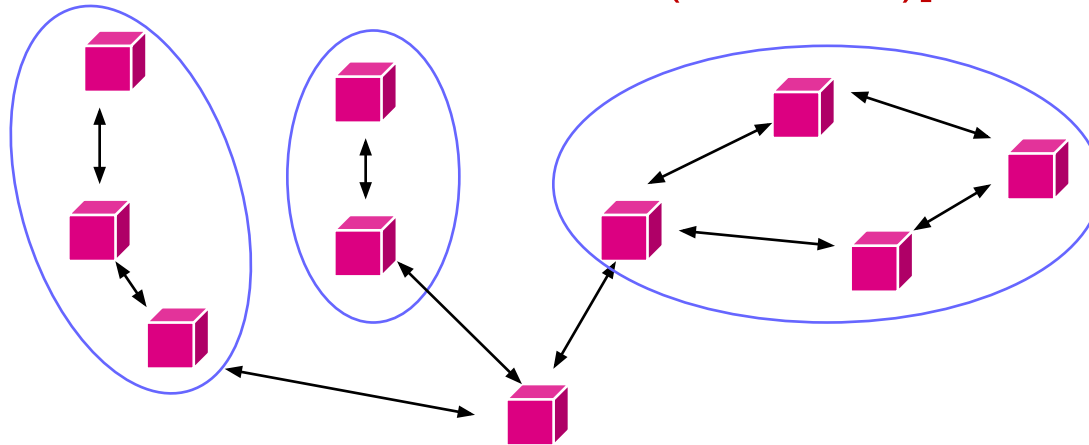
(Structured / Semi-Structured / Un-Structed :: Depending on Application Nature)



Multi-hop Infrastructure-less: Wireless Sensor Networks

Energy is the primary constraint

Low power Wireless Sensor Network (Low PAN)[IEEE 802.15]

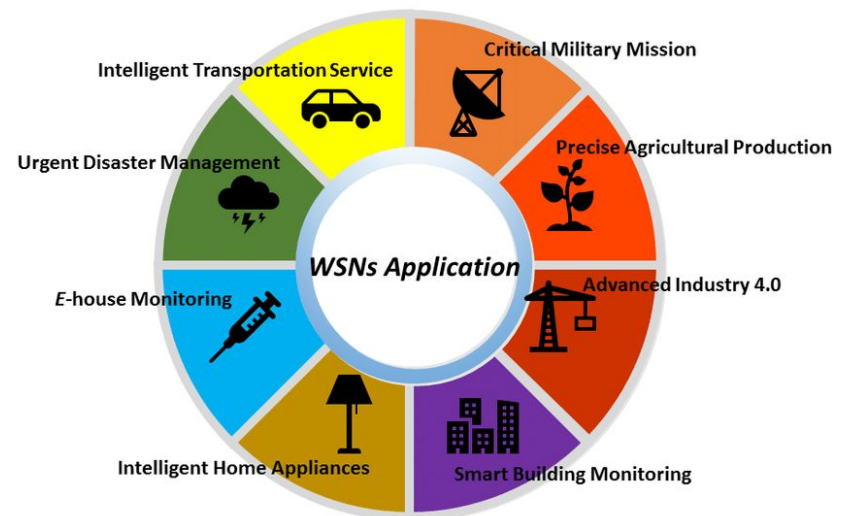
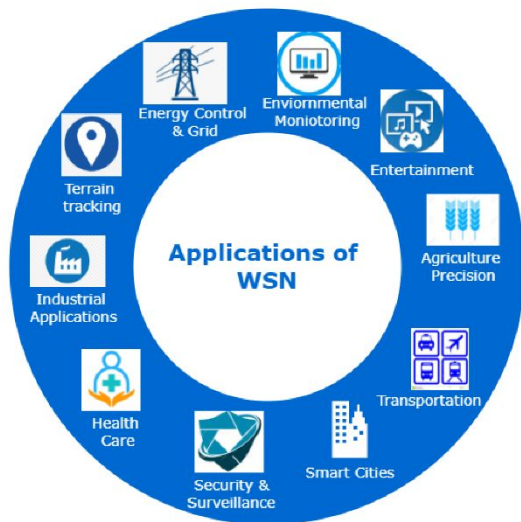


- Nodes powered by non rechargeable batteries
- Data flows usually to centralized location.
- Low per-node rates but up to 100,000 nodes.
- Data highly correlated in time and space.
- Nodes can cooperate in transmission, reception, compression, and signal processing.

Sensor Networks

- Short-range networks must consider transmit, circuit, and processing energy.
 - Sophisticated techniques not necessarily energy-efficient.
 - Sleep modes save energy but complicate networking.
- Changes **everything** about the network design:
 - Delay vs. throughput vs. node/network lifetime tradeoffs.
 - Optimization of node cooperation.

Sensor Networks :: Applications



Sensor Networks :: Applications

Constant monitoring & detection of specific events

Military, battlefield surveillance

Forest fire & flood detection

Habitat exploration of animals

Patient monitoring

Home appliances

Sensor Networks :: Applications

Precision Agriculture: (Structured / Semi-Structured)

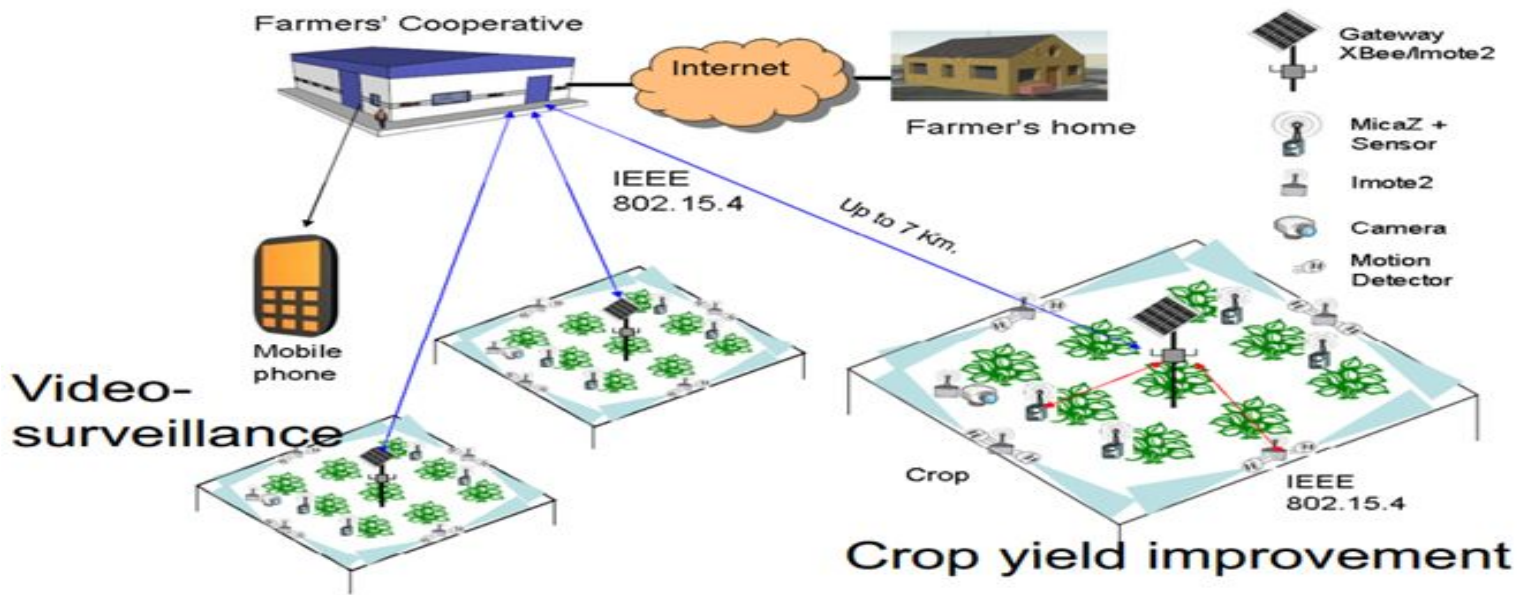


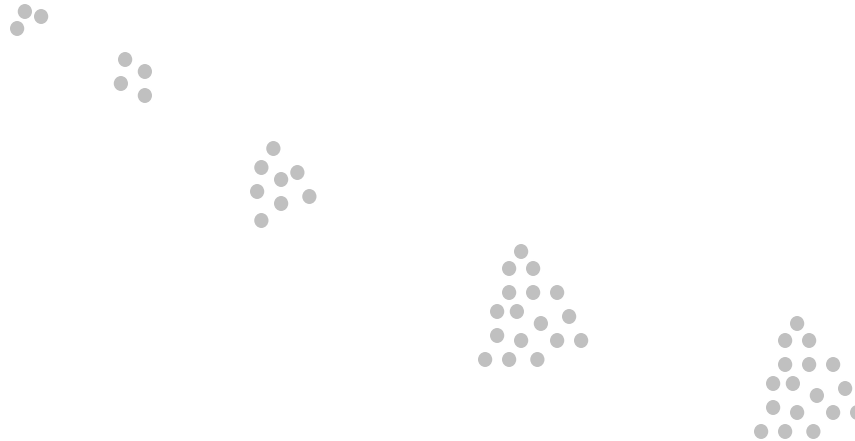
Figure: An example of Low PAN Application

Sensor Networks :: Applications

Military: (Un-Structured)



tactical monitoring



Sensor Networks :: Applications

Smart Home/Smart Office: (Structured)

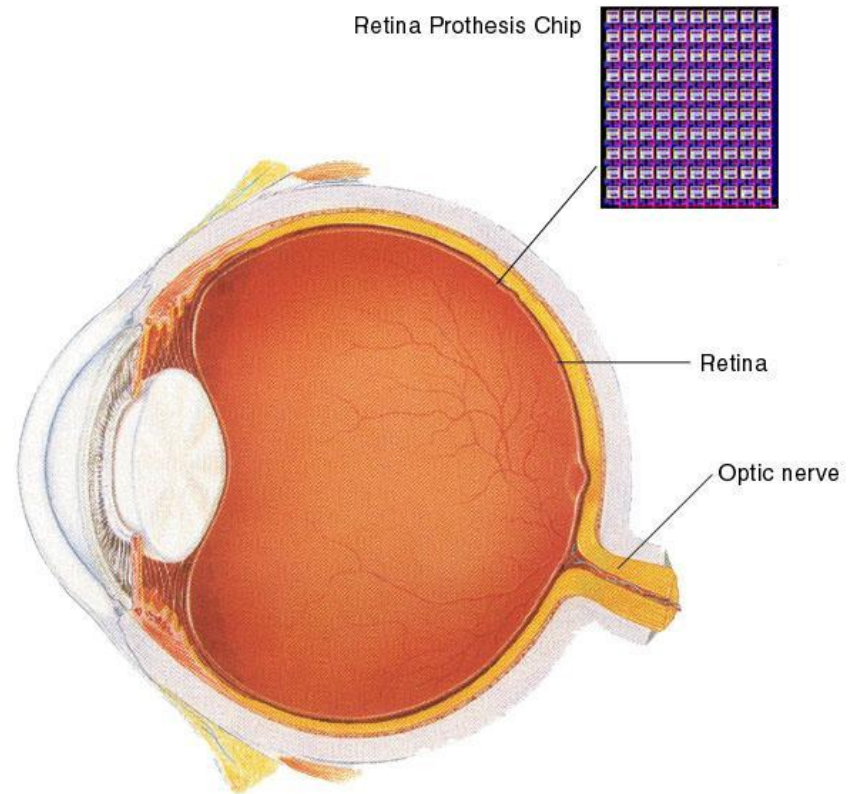


- Sensors controlling appliances and electrical devices in the house
- Better lighting and heating in office buildings

Sensor Networks :: Applications

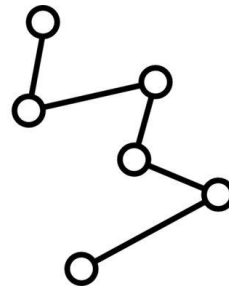
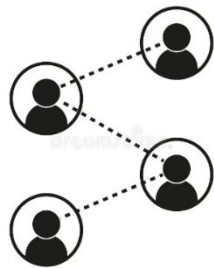
Medical/Biomedical: Body Area Network: (Structured)

- Health Monitors
 - Glucose
 - Heart rate
 - Cancer detection
- Chronic Diseases
 - Artificial retina
 - Cochlear implants
- Hospital Sensors
 - Monitor vital signs
 - Record anomalies



Delay/Disruption Tolerant Network (DTN)

(Un-Structured)



Delay Tolerant Network (DTN)

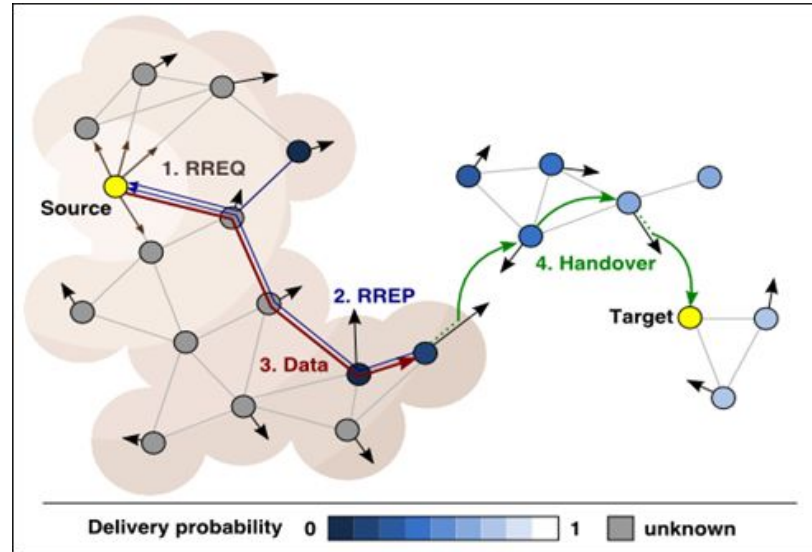


Figure:

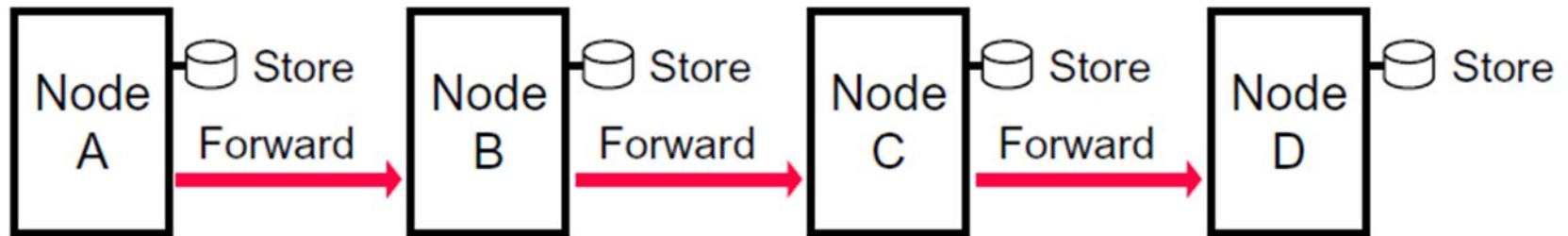


Figure: Store-And-Forward Message Switching

Delay Tolerant Network

- DTN architecture expects nodes to store bundles for an extended period.
- Whenever possible, each received data packet is forwarded immediately.
- If forwarding is not currently possible but is expected to be possible in the future, the packet is stored for future transmission.

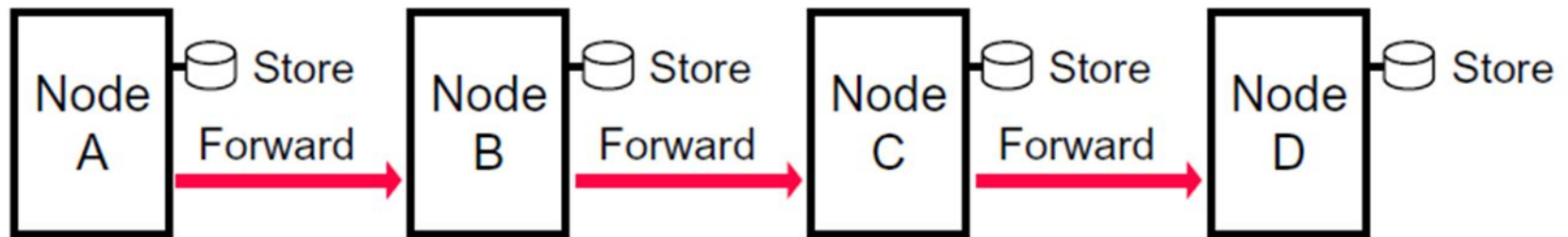


Figure: Store-And-Forward Message Switching

Delay Tolerant Network

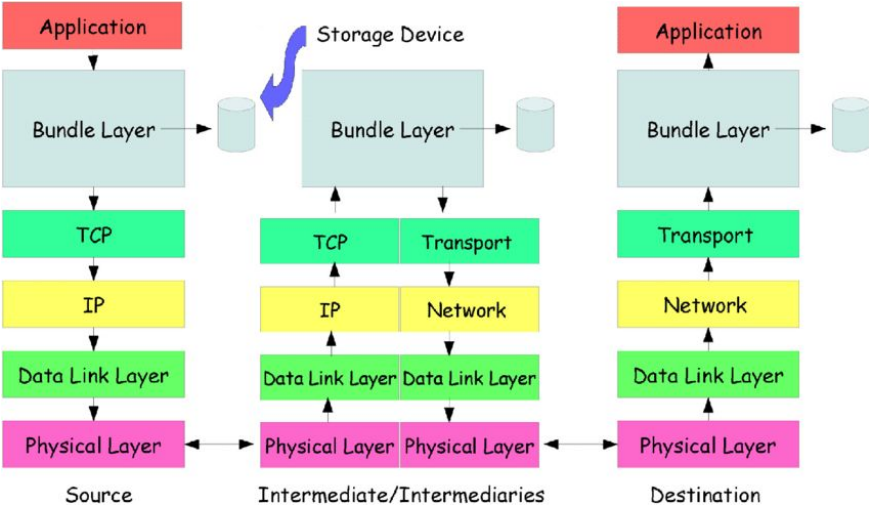


Figure:

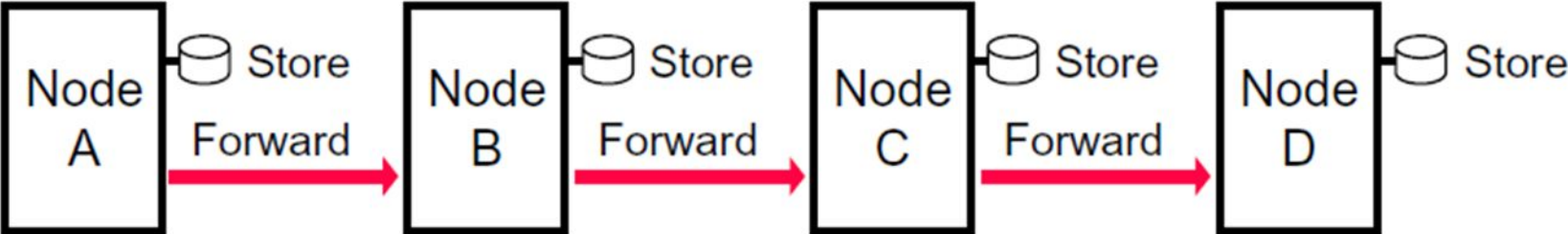


Figure: Store-And-Forward Message Switching

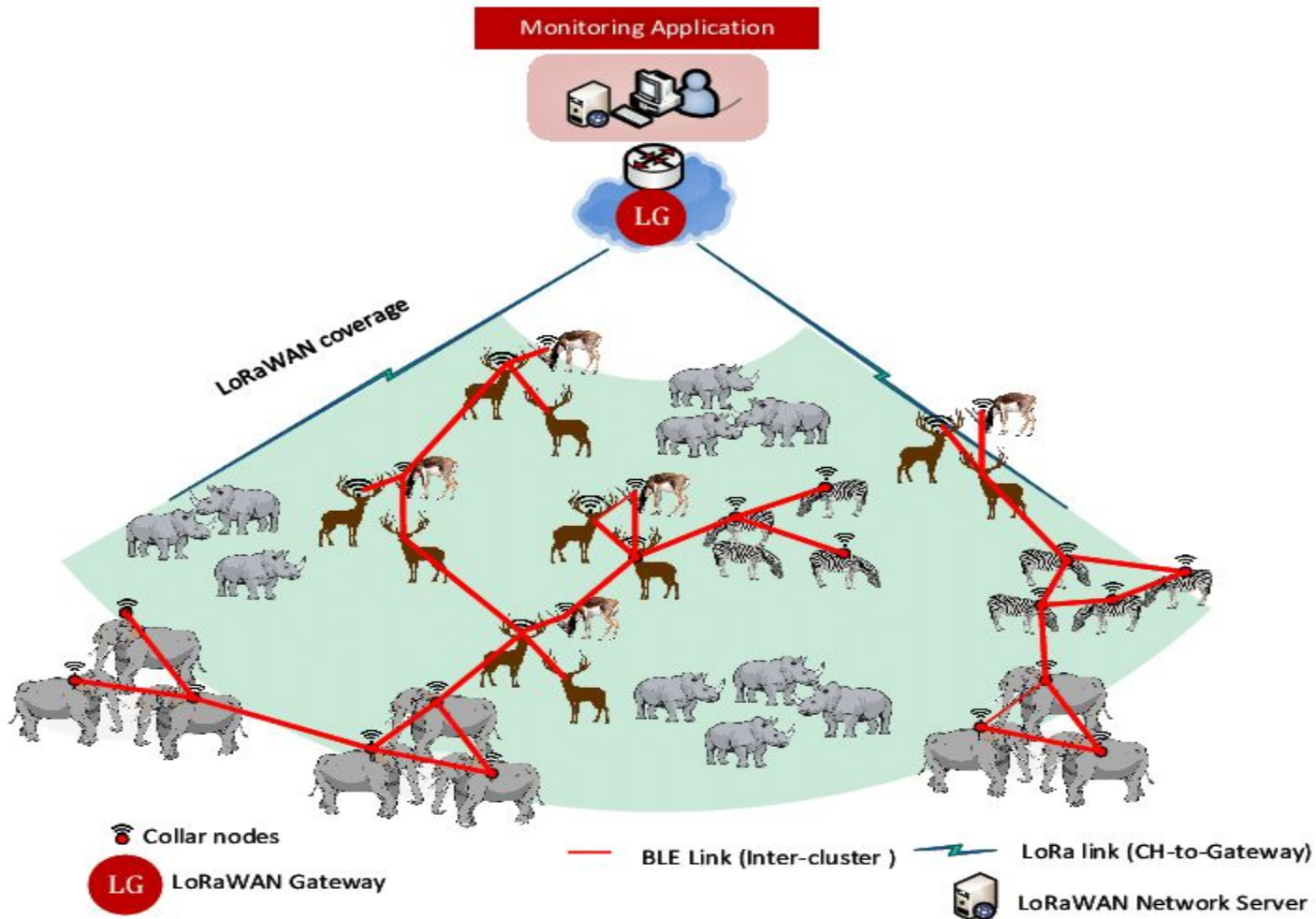
Delay Tolerant Network (DTN):: Applications

Mobile-enabled delay tolerant networking in rural developing regions



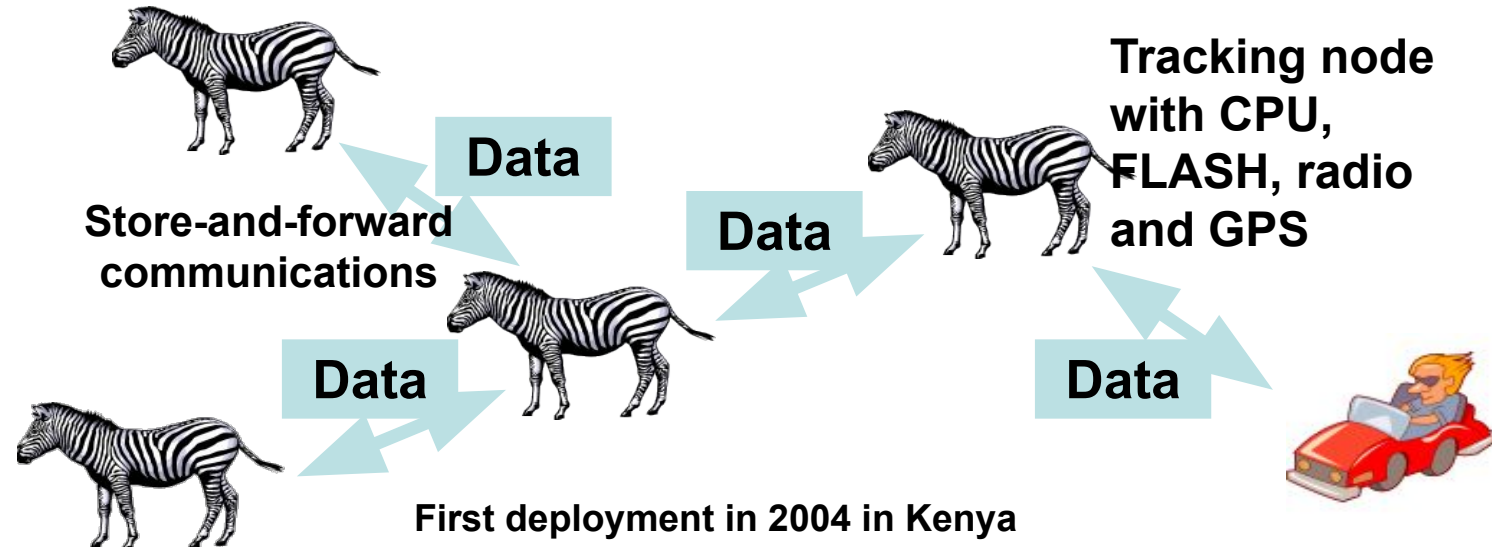
Fig. 1. A delay-tolerant network between the city of Durban and

Delay Tolerant Network (DTN):: Applications



Delay Tolerant Network (DTN):: Applications

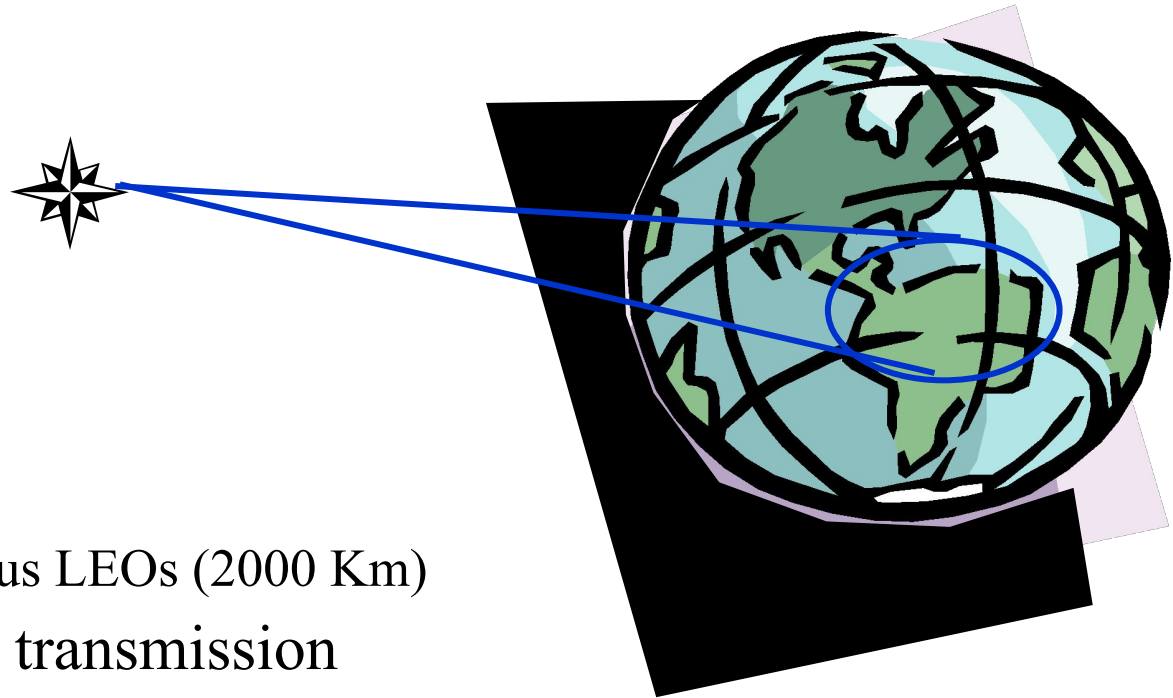
ZebraNet (a real life application)



Delay Tolerant Network (DTN):: Applications

Satellite Systems

- Cover very large areas
- Different orbit heights
 - GEOs (39000 Km) versus LEOs (2000 Km)
- Optimized for one-way transmission
 - Radio (XM, DAB) and movie (SatTV) broadcasting



DTN :: Characteristics

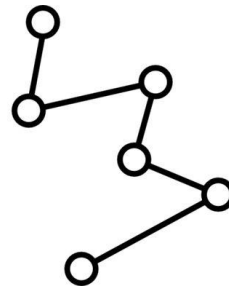
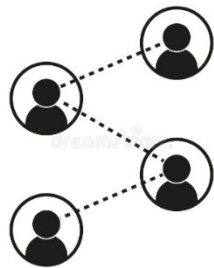
- Internet environment
 - End-to-end RTT is not large.
 - Some path exists between endpoints.
 - E2E reliability using ARQ works well.
 - Packet-switching is the right abstraction.
- DTN characteristics
 - Very large delays.
 - Intermittent and scheduled links.
 - Different network architectures.
 - Conversational protocols fail.
 - No ARQ.

DTN :: Challenges

- Applications assume e2e ‘connections’
- Routing does not understand schedules
- Long-term disconnection causes failures
- End-to-end reliability is poor with high loss rates
- E2E performance is poor with high delays

Vehicular Ad-hoc Networks

(Un-Structured)



Vehicular Ad-hoc Networks

VANET

- a form of Mobile ad-hoc network
- provide communication
 - among nearby vehicles
 - nearby fixed equipment

Vehicular Ad-hoc Networks

How vehicular communications work:

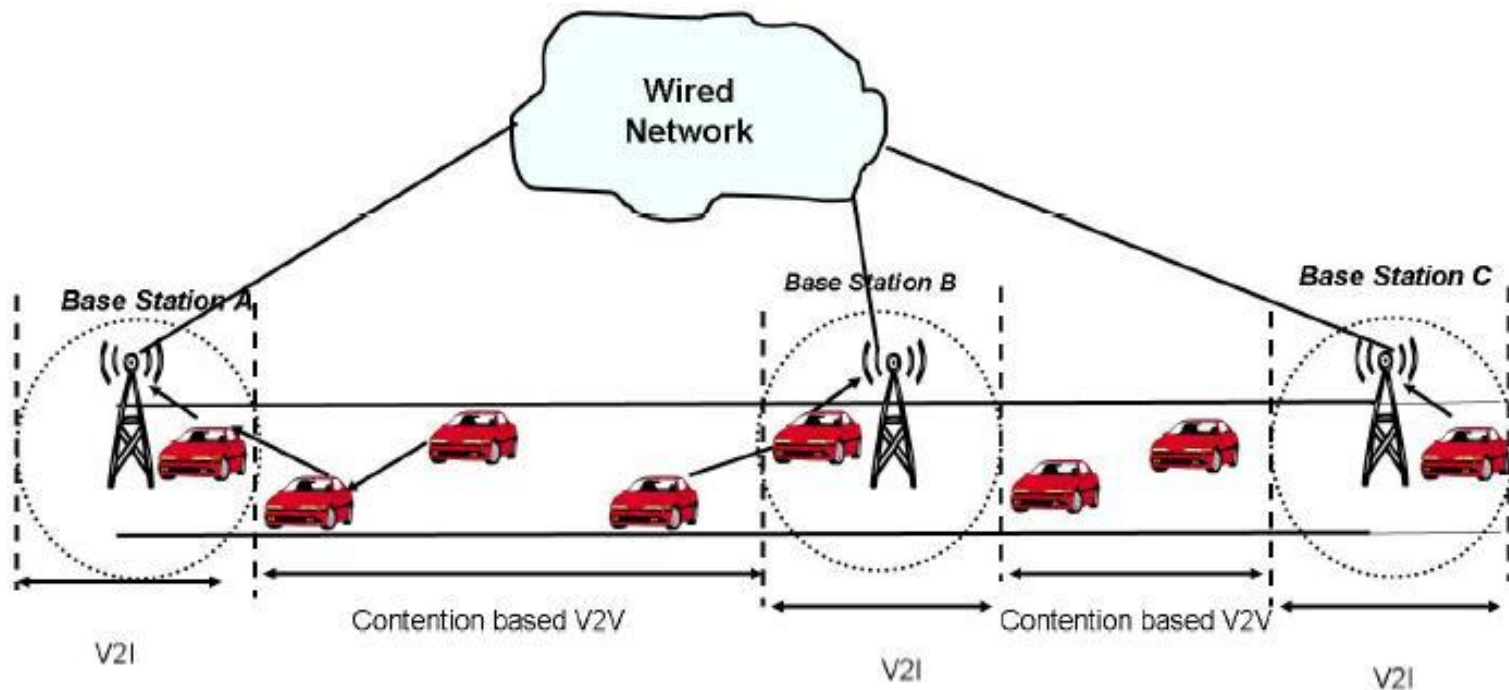
Road-Side infrastructure Units (RSUs), are equipped with **on-board processing and wireless communication modules**



Vehicular Ad-hoc Networks

How vehicular communications work:

- Vehicle-to-Vehicle (*V2V*)
- Vehicle-to-Infrastructure (*V2I*) communication will be possible



Car-to-X (C2X) Communication Patterns

- Vehicle-to-X (V2X)/ Car-to-X (C2X),
- Inter-Vehicle Communication (IVC),
- Vehicular ad-hoc network (VANET),
- ...

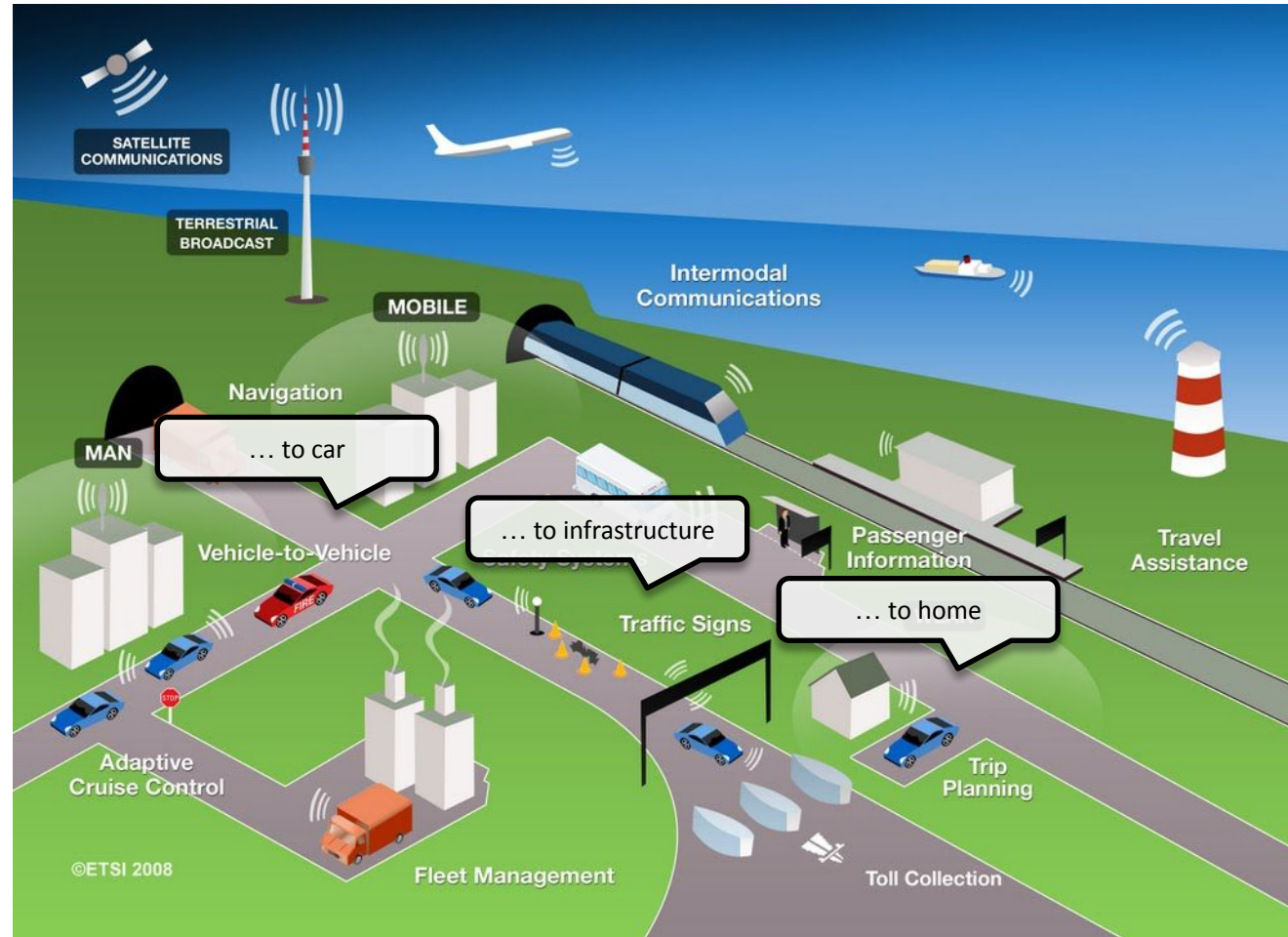
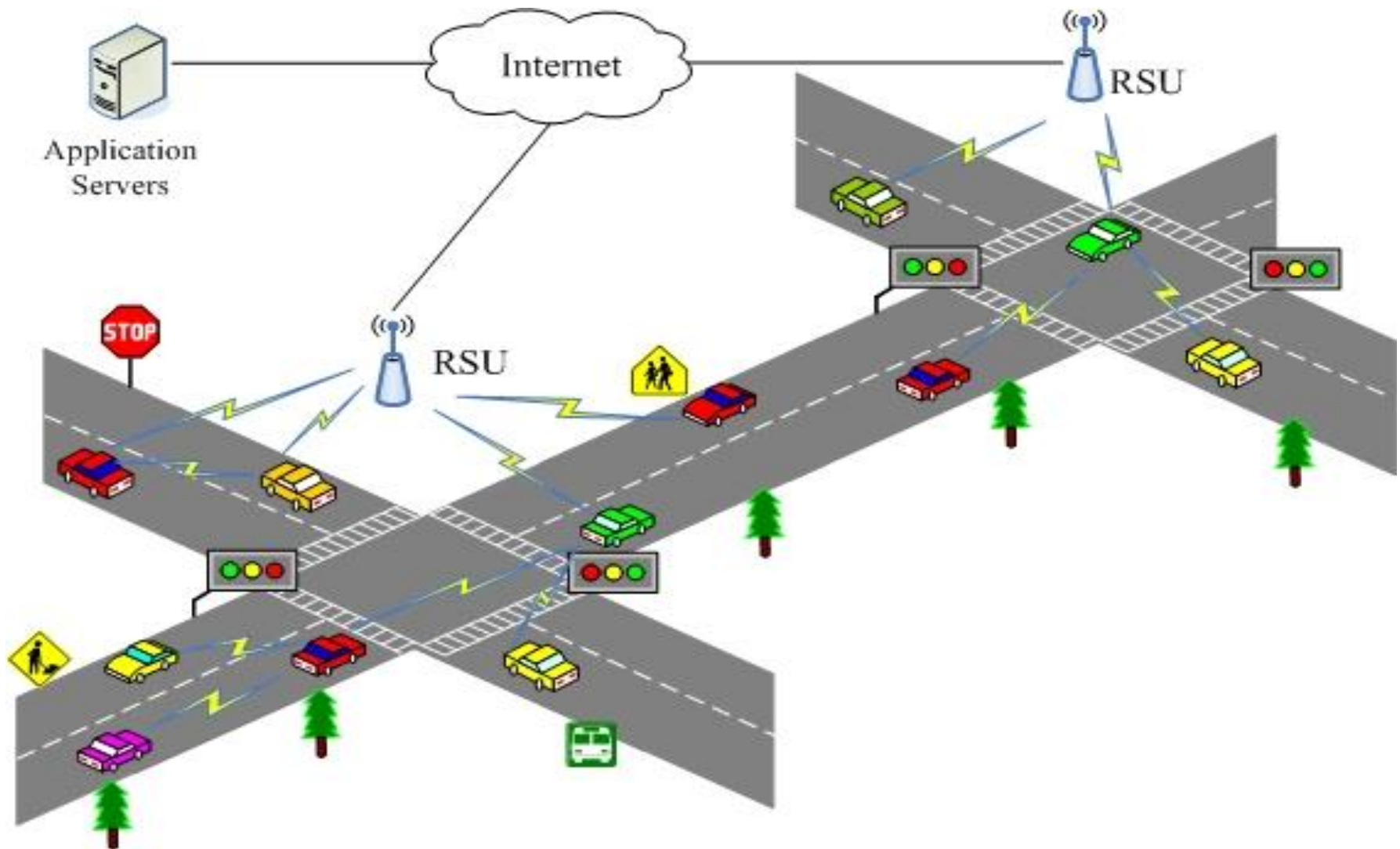


Illustration: ETSI

Car-to-X (C2X) Communication



Intelligent Transportation Systems (ITS)



Intelligent Transportation Systems (ITS)



Intelligent Transportation Systems (ITS)

What can ITS provide ?

- Safety
 - Efficiency
 - Traffic and road conditions
 - Road signal alarm
 - Local information
-

Intelligent Transportation Systems (ITS)

Few Emerging Applications that ITS provide ?

- Autonomous Intersection Management (AIM)
 - Vehicle Platooning for Fuel and Space Efficiency
 - Eco-Driving and Green Routing
 - Cooperative Perception and Safety Sharing
 - Autonomous On-Demand Public Transit (Dynamic ride-pooling)
 - Automated Smart Parking
 - Real-Time Traffic Prediction and Control
 - Cooperative Highway Merging and Lane Changing
 - Dynamic Speed and Lane Recommendations
-

Car-to-X (C2X) Communication

- Autonomous Intersection Management (AIM):



However, future AIM will adhere Strictly to Fundamental Traffic Rules,
Prioritizing Safety as the Top Priority without Human Intervention

Car-to-X (C2X) Communication

■ Autonomous Intersection Management (AIM):

- Replaces traffic lights with autonomous intersection control, where Connected and Autonomous Vehicles (CAVs) communicate with a central coordinator to avoid collisions.

Benefits:

- Minimal vehicle waiting time
- Reduced fuel consumption and emissions
- Elimination of signal delays
- Improved intersection throughput
- Enhanced safety

Research Extensions:

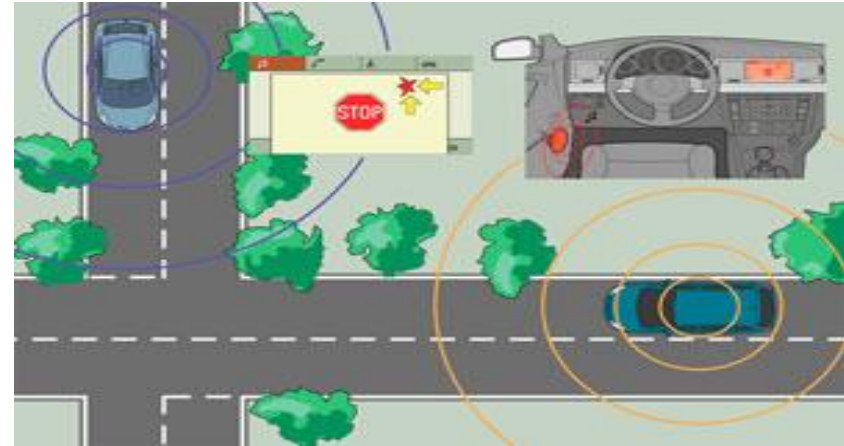
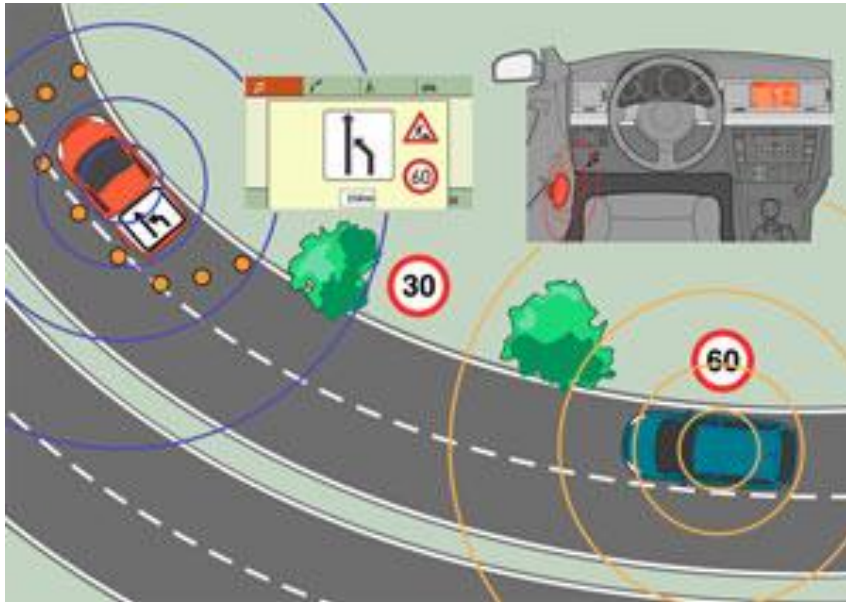
- Robustness to Communication Delays and Packet Loss
- Human-Driven Vehicle (HDV) and CAV Coexistence Models
- Integration with Eco-Driving Strategies
- Scalability in High-Density Urban Grids



Car-to-X (C2X) Communication

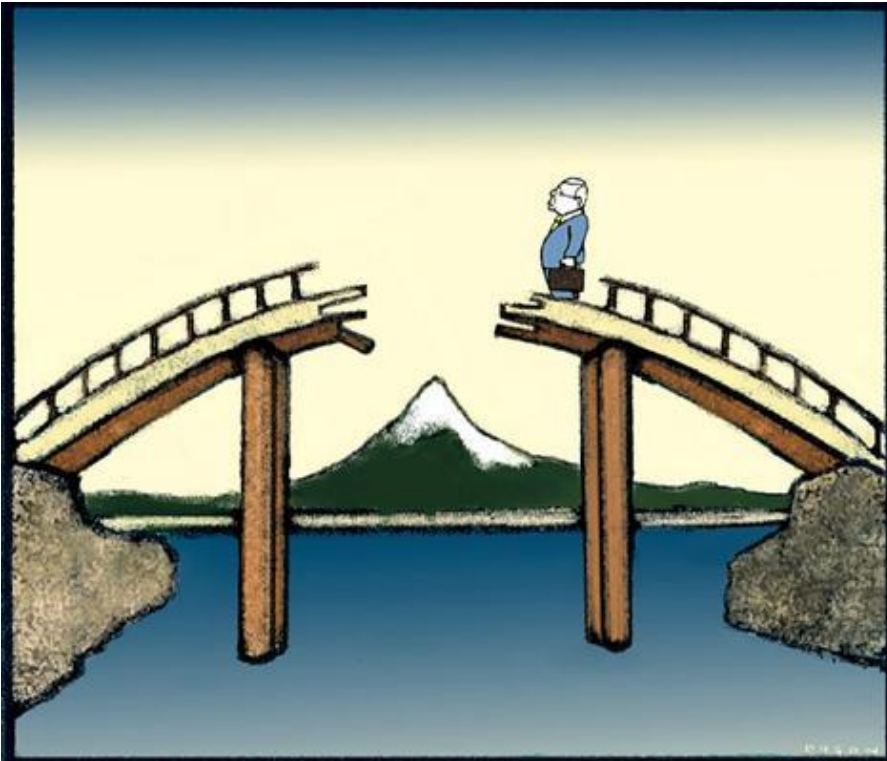
What can Car-to-X (C2X) communication provide ?

**Cooperative Perception and Safety Sharing
Warning!!**



Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

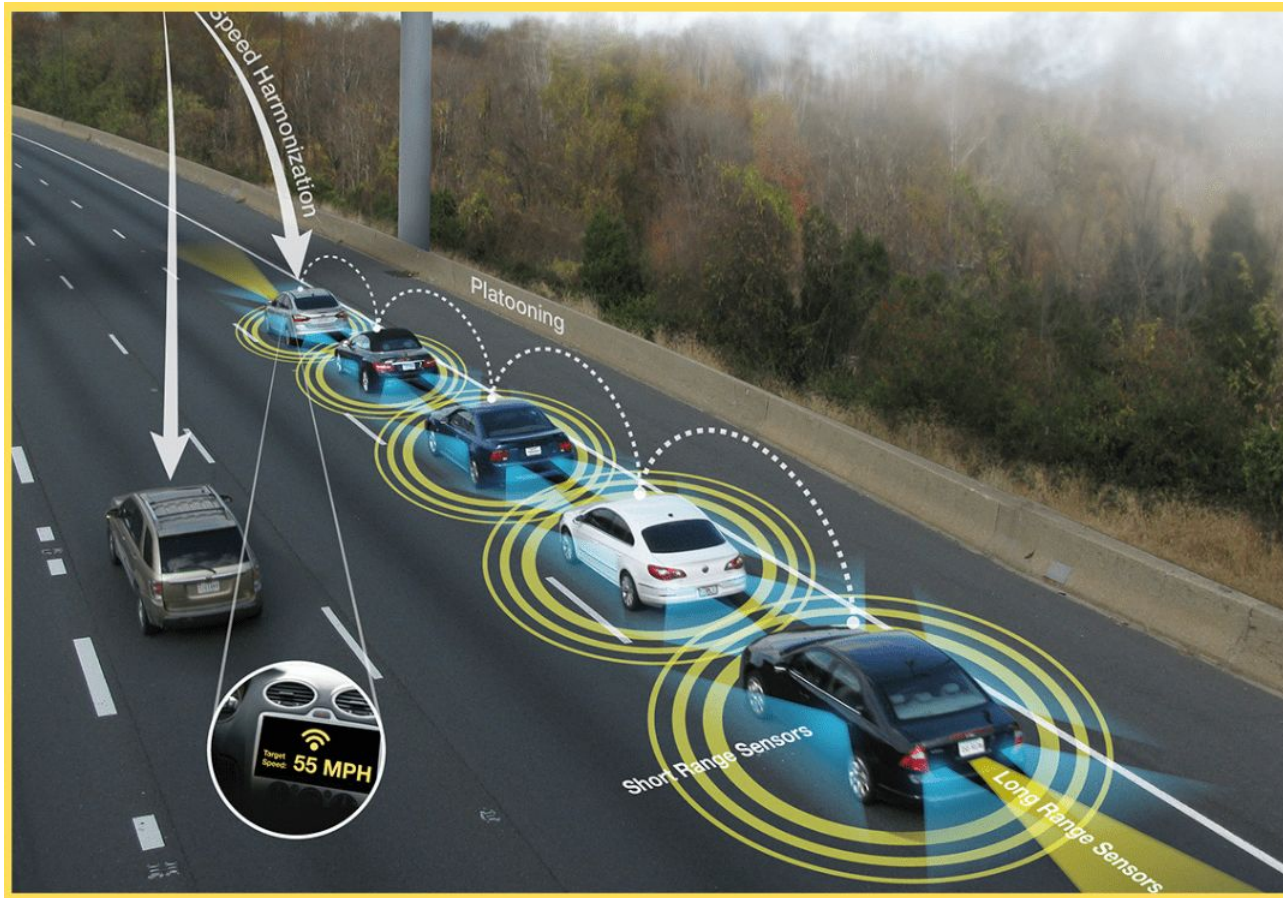


Warnings!!!
Incident Detection and Reporting

Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Vehicle Platooning :: Eco-Driving



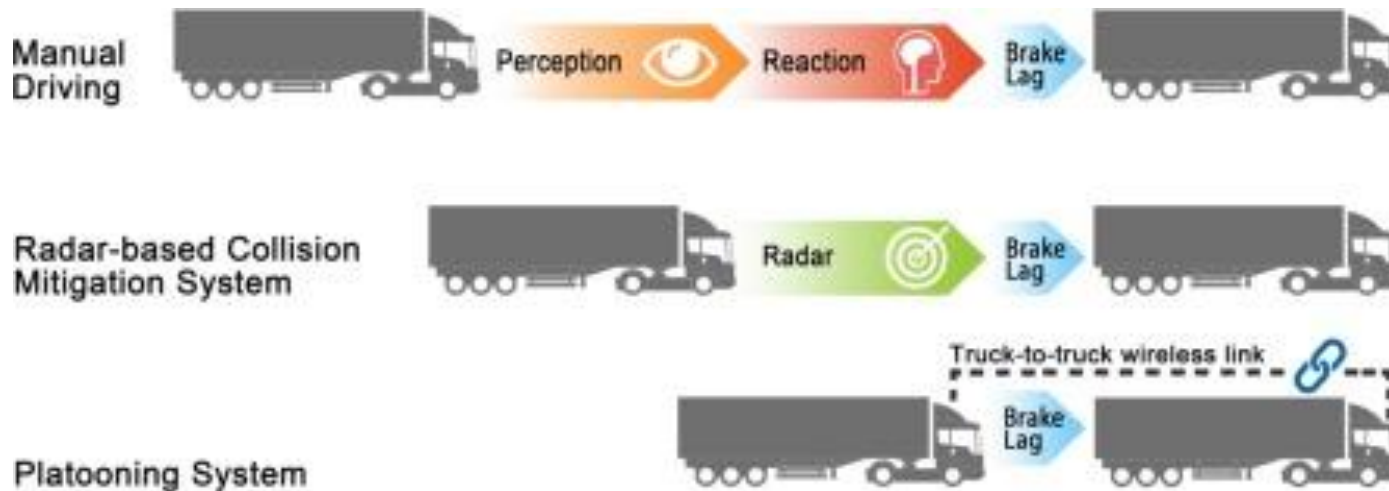
Vehicle-to-Vehicle (V2V) Communication

Vehicle-to-Vehicle (V2V) communication enables direct wireless data exchange between vehicles, allowing them to share real-time information such as position, speed, and braking status.

Vehicle Platooning:: Eco-Driving

Faster and more synchronized braking response

- enhancing safety
- enabling closer following distances
- improving traffic flow efficiency



Vehicle-to-Vehicle (V2V) Communication

Manual Driving:

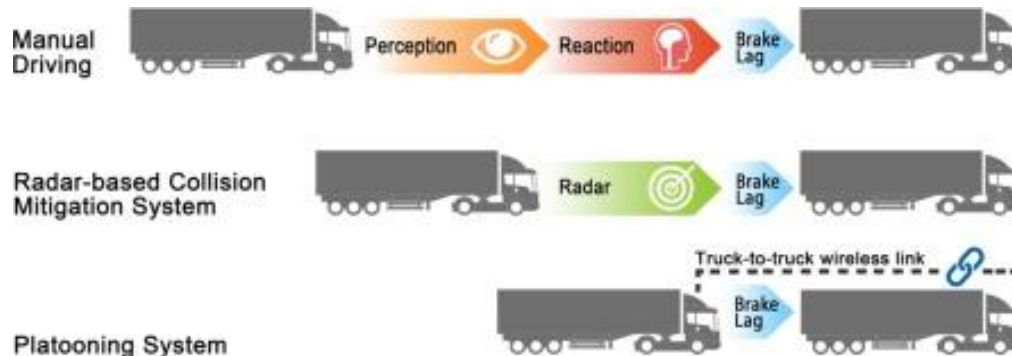
- Depends on **human perception and reaction**.
- Causes **significant brake lag** and larger following distances.

Radar-Based Systems:

- Uses sensors to detect deceleration.
- Reduces reaction time but still has **processing delays**.

V2V-Based Platooning:

- Shares braking information **wirelessly and instantly**.
- Enables **simultaneous braking**, minimizing brake lag and allowing **closer spacing**.

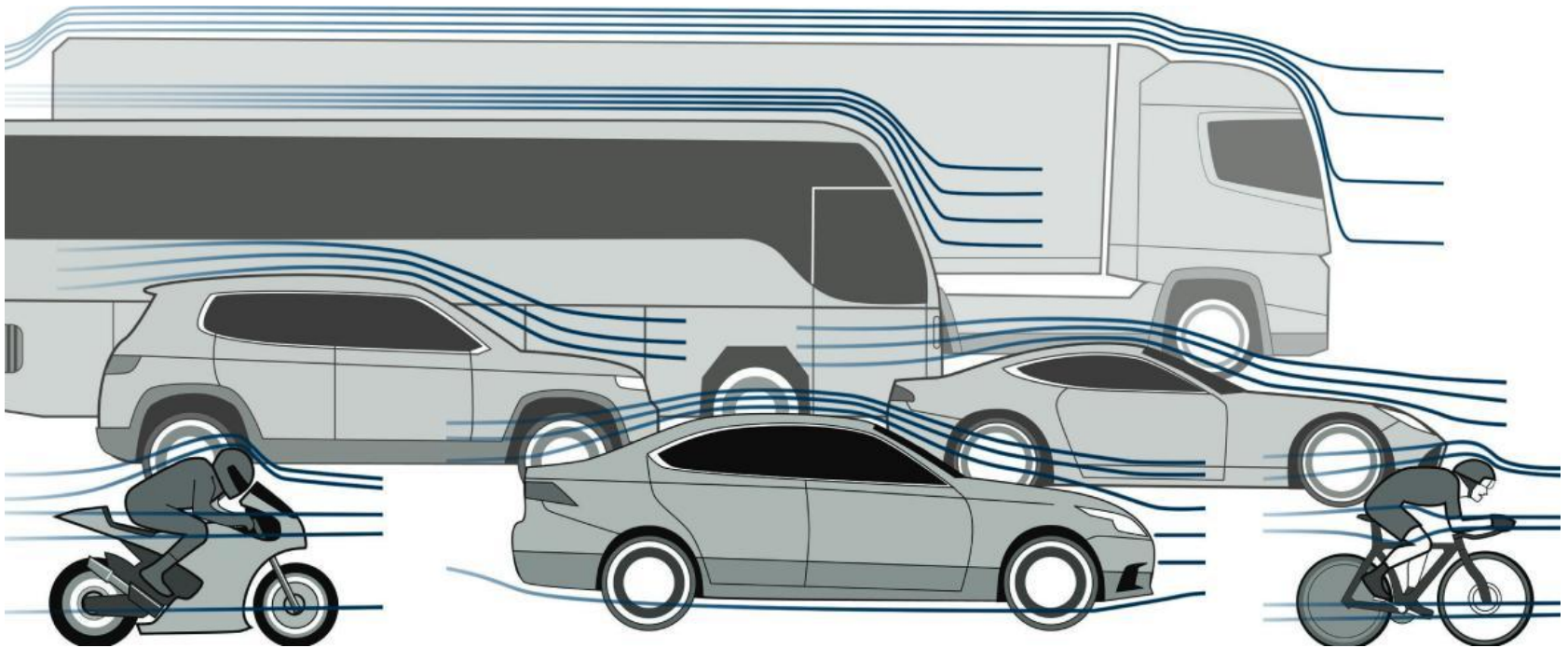


Vehicle-to-Vehicle (V2V) communication enhances emergency braking in truck platooning by enabling **faster and synchronized braking responses** across all vehicles, improving both **safety** and **traffic flow efficiency** compared to traditional systems.

Vehicle-to-Vehicle (V2V) Communication

What can V2V communication provide ?

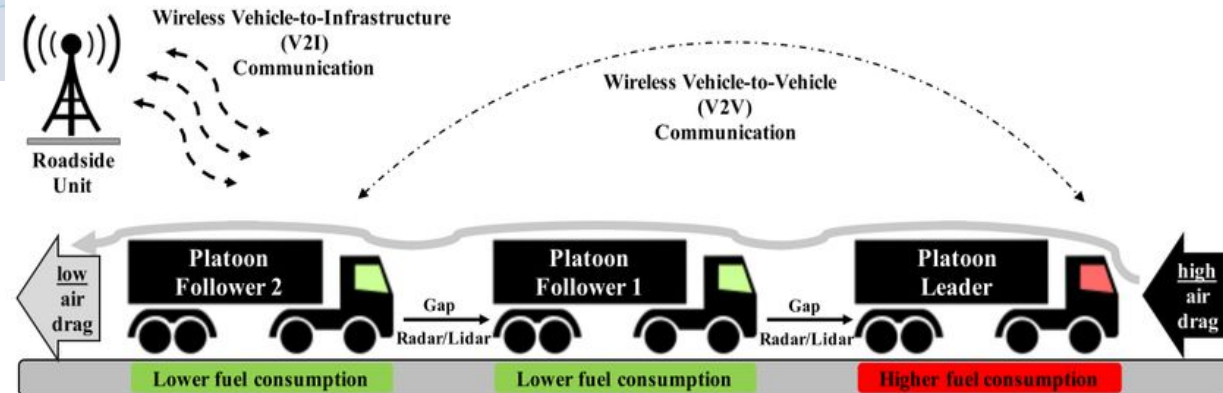
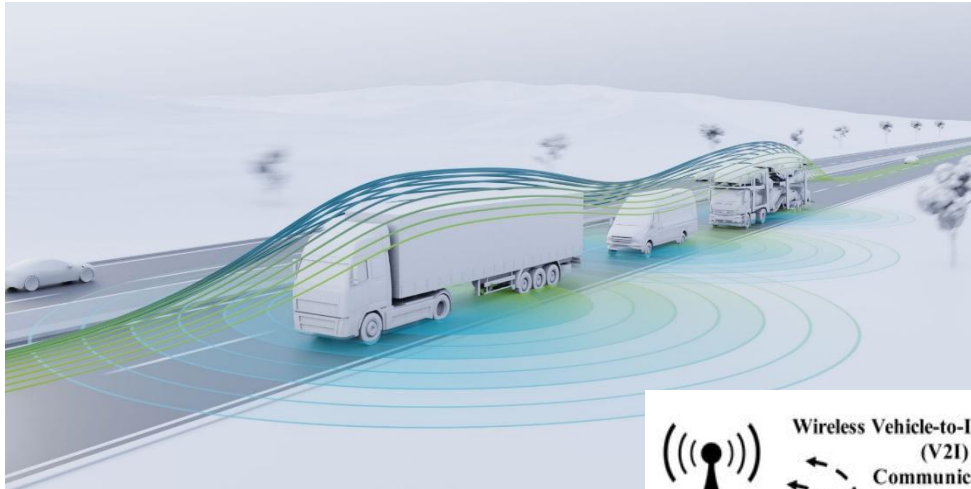
Vehicle Platooning for Improved Fuel Efficiency
through **Reduced Aerodynamic Drag**



Vehicle-to-Vehicle (V2V) Communication

What can V2V communication provide ?

Vehicle Platooning for Improved Fuel Efficiency
through **Reduced Aerodynamic Drag**



Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

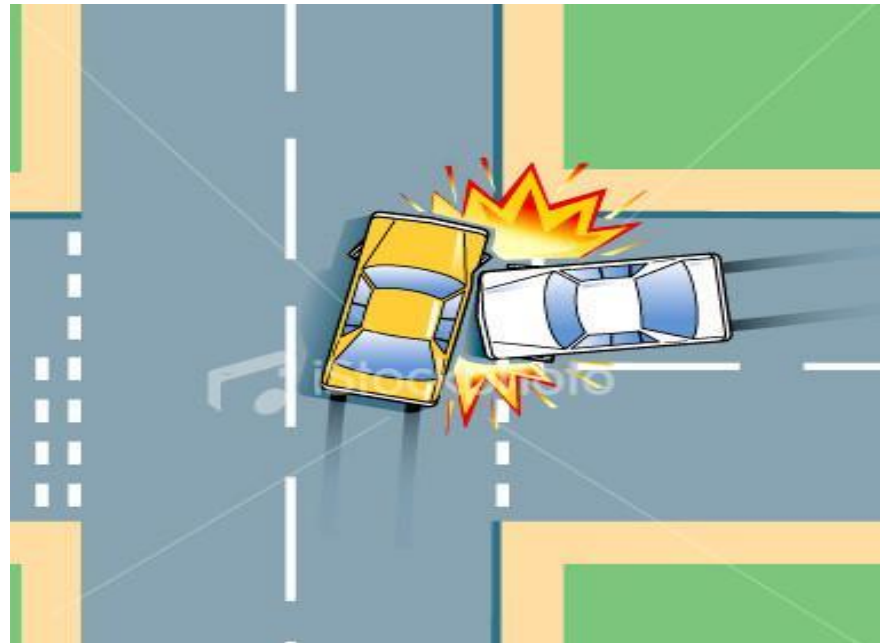
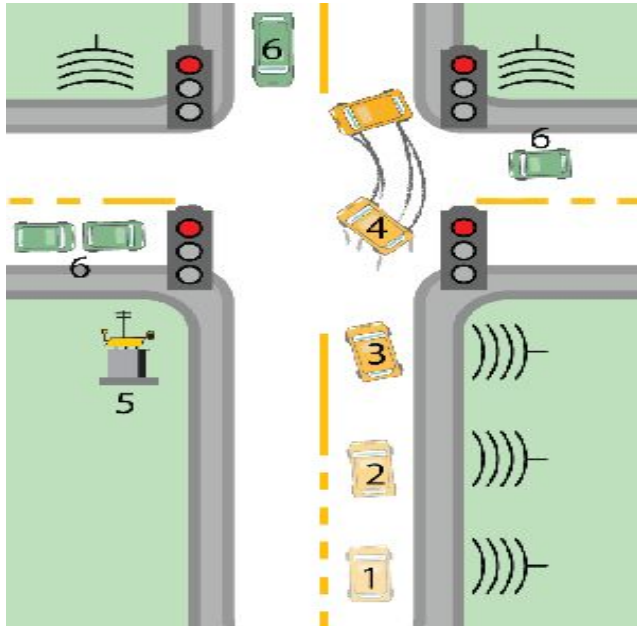
Intelligent Traffic Flow Management



Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Road Traffic Accidents Avoidance

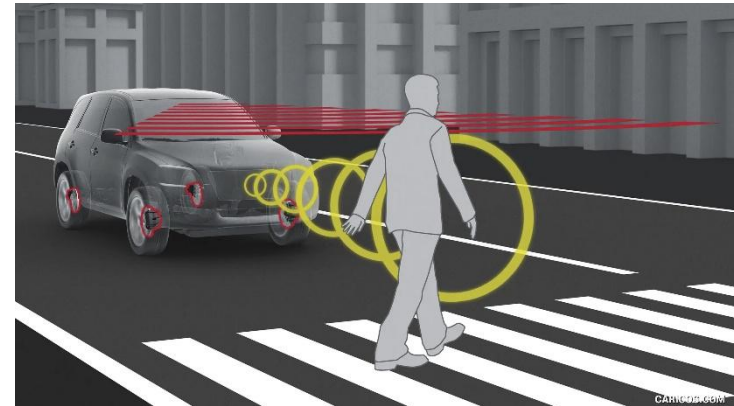


Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?



Pedestrian and Cyclist Protection



Challenges of C2X Communication

Communication

- Highly varying channel conditions
- High congestion, contention, interference
- Tightly limited channel capacity

Networking

- Unidirectional Links
- Multi-Radio / Multi-Network
- Heterogeneous equipment

Mobility

- Highly dynamic topology
- But: predictable mobility
- Heterogeneous environment

Security

- No (or no reliable) uplink to central infrastructure
- Ensuring privacy
- Heterogeneous user base

Thank You